HG-SERIES

LOW-VOLTAGE AC POWER CIRCUIT BREAKER

Instruction Manual



Type covered in this manual

- HGL Type : HGL08, HGL16, HGL20, HGL25, HGL32, HGL40, HGL60

Notice

- Read this instruction manual carefully before using this LOW-VOLTAGE AC POWER CIRCUIT BREAKER.
- This instruction manual does not include all information regarding installation and maintenance.
- For further information, please contact HD Hyundai Electric or your local dealer.

HD HYUNDAI ELECTRIC

For Operational Safety of Air Circuit Breakers

Safety Practices

This instruction manual applies only to HG-Series LOW-VOLTAGE AC POWER CIRCUIT BREAKER (ACB) regarding installation and maintenance procedures. Installing and maintaining these products improperly may result in serious personal injury, property damage, or even death. Therefore this instruction manual must be read and understood at any step in unpacking, assembly, operation, and maintenance of the breaker. Only qualified persons who are familiar with installing and maintaining circuit breakers are permitted to work on breakers, and this instruction manual should be accessible to those persons at any time.

Signal Words

Signal words used in this instruction manual are divided by DANGER, WARNING, and CAUTION depending on the situation.

⚠ DANGER — Indicates an imminently hazardous point that, if ignored, a serious accident causing death or severe injury could occur. This is the most dangerous point.

⚠ WARNING — Indicates a potentially hazardous point that, if ignored, a serious accident causing death or severe injury could occur.

⚠ CAUTION — Indicates a potentially hazardous point that, if ignored, minor or moderate injury could occur. This signal reminds operators to work safely.

Transportation Precaution

⚠ DANGER

Never lift the ACB over an area where there are people.
Never stand under the ACB.

OCR Field Test Precautions

⚠ CAUTION

- OCR field tests and setting changes must be performed by qualified personnel.
- After completion of OCR tests, make sure to return the original values.
- Failure to do may cause a fire or burnout.

Operation Precautions

⚠ DANGER

- Never touch live terminal parts.
  There is a risk of electric shock.
- Do not leave the ACB in the draw-out position.
  Dropping the ACB cause serious injury.

⚠ CAUTION

- If the ACB trips open automatically, remove the cause before closing the ACB.
  Otherwise, a fire could occur.
- For the ACB with the fixing block, be sure to loosen the block screws before drawing out the ACB.
  Otherwise, damage to the ACB may occur.

Installation Precautions

⚠ CAUTION

- Installation work must be performed by qualified personnel.
- Prior to commencing any installation work, open an upstream circuit breaker to shut off all sources of power.
  Otherwise, electric shock may occur.
- Tighten terminal bolt up securely to the specified torque.
  Otherwise, a fire may occur.
- Fix the draw-out cradle of the ACB firmly on a flat level surface using mounting screws.
  Otherwise, draw-out operation may cause the ACB to fall.
- To ensure adequate arc space (insulation distance), avoid blocking the arc gas vents of the ACB.
  Otherwise a burn may result from high temperature arc gas.
- Do not place the ACB in such as an area that is subject to high temperature, high humidity, dusty air, corrosive gas, vibration and shock, or other unusual conditions.
  Mounting in such area could cause a fire, non-tripping, or malfunction.
- Install the ACB to prevent dust, concrete powder, iron powder, and rainwater from entering the unit.
  These materials could cause a fire or non-tripping.
- For the ACB with four poles, connect the neutral wire of a 3-phase, 4-wire cable to the N-phase pole (on the right end).
  Otherwise, an overcurrent may cause non-tripping resulting a fire.

Maintenance and Inspection Precautions

⚠ CAUTION

- ACB maintenance, inspection and/or parts replacement must be performed by qualified personnel.
- Prior to commencing any work on the ACB, open an upstream breaker or the like to isolate all sources of power/voltage from both the primary and auxiliary circuits.
  Otherwise, electric shock may result.
- Prior to commencing ACB internal inspection, make sure that the ACB is open and the closing spring is released.
  Otherwise, fingers or tools could be pinched in the internal mechanism, resulting injury.
- Retighten the terminal screws periodically to the specified torque.
  Otherwise, a fire could result.
- Be sure to reinstall the arc chamber if removed.
  Failure to do so or incorrect installation of the arc chamber may result in a fire or burn.
- Do not touch ACB live parts (contacts in particular), or structural parts close to a live part immediately after opening the ACB to stop supplying power.
  Otherwise, remaining heat may cause burn or residual charged voltage may cause electric shock.
- Do not bring your hand or face close to the arc gas vent of the arc chute while the ACB is closed.
  Otherwise, a burn may result from high-temperature arc gas blowing out of the arc gas vent when the ACB trips open.

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01 Receiving and Handling

Upon receipt of your breaker, check the following first.
HE air circuit breakers are completely assembled, inspected and tested both electrically and mechanically at the factory, then shipped in fully guaranteed condition in construction and operation.

1.1 Storage

While it is recommended that the breaker be used as soon as you have received it. If it is necessary to store the ACB, note the followings.

1) Store the breaker in a dry indoor location to prevent condensation due to sudden changes in ambient temperature, which is quite harmful to the breaker insulation.
2) Store the breaker in a clean place free of corrosive gases, dirt, dust and salinity (NaCl).
   In particular, a mixture of cement dust and moisture may cause corrosion damage to metal parts of the ACB.
   Fully protect the breaker from such mixtures.
3) Place the breaker on a flat, level surface in its normal position.
4) Do not place the breaker directly on the floor.
5) When store the ACB without electronic circuit : -25℃ - 85℃
   When store the ACB with electrofin circuit : -15℃ - 70℃

1.2 Transportation

When transporting the breaker, follow these instructions.

1) When lifting up the breaker, suspend wire rope from lifting lug and take care to ensure that the wire ropes do not touch the arc chute and protection relay. When lifting up the breaker, be sure to lift it up slowly.
2) Lower the breaker onto a flat surface.
3) Avoid impacts and shocks to the breaker during transportation.

01 Receiving and Handling

1.3 Enclosure Size

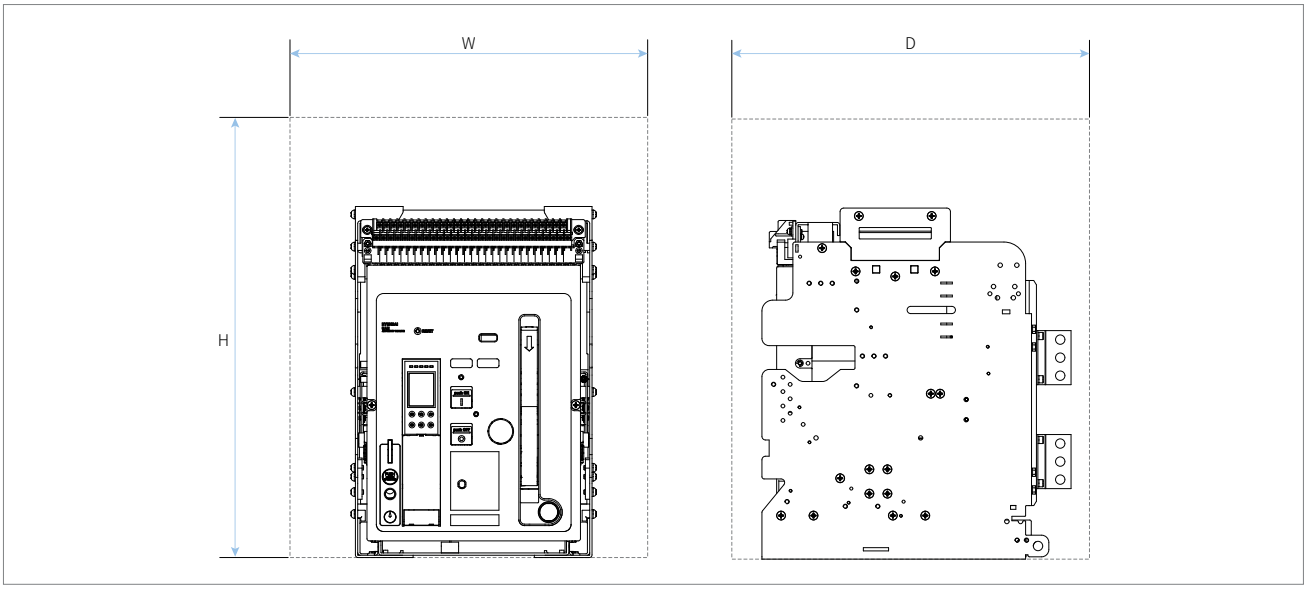
1) Minimum enclosure size

Enclosure Dimension			H	W	D
A Frame	3P	In	19.69	21.18	15.00
		mm	500	538	381
	4P	In	19.69	24.53	15.00
		mm	500	623	381
B, Q Frame	3P	In	19.69	24.02	15.00
		mm	500	610	381
	4P	In	19.69	28.50	15.00
		mm	500	725	381
C Frame	3P	In	20.47	32.83	15.00
		mm	520	834	381
	4P	In	20.47	39.53	15.00
		mm	520	1004	381
D Frame	3P	In	20.47	38.43	15.00
		mm	520	976	381
	4P	In	20.47	47.48	15.00
		mm	520	1206	381

※ - Enclosure dimension is followed by UL/ANSI Standard.

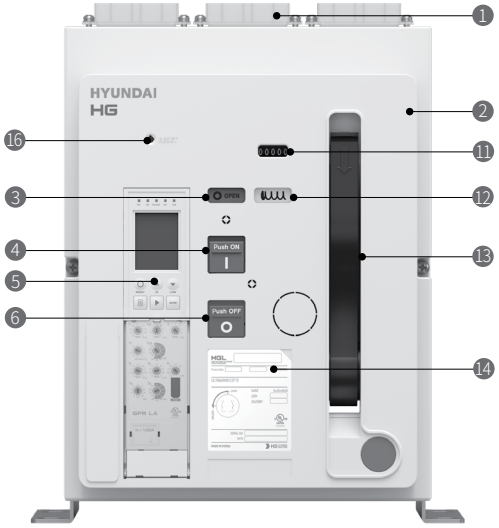
1) Minimum enclosure ventilation size

Enclosure Dimension		Top mm^2	Bottom mm^2
A Frame	3P	12,000	12,000
	4P	15,000	15,000
B, Q Frame	3P	35,000	35,000
	4P	45,000	45,000
C Frame	3P	50,000	50,000
	4P	50,000	50,000
D Frame	3P	50,000	50,000
	4P	50,000	50,000

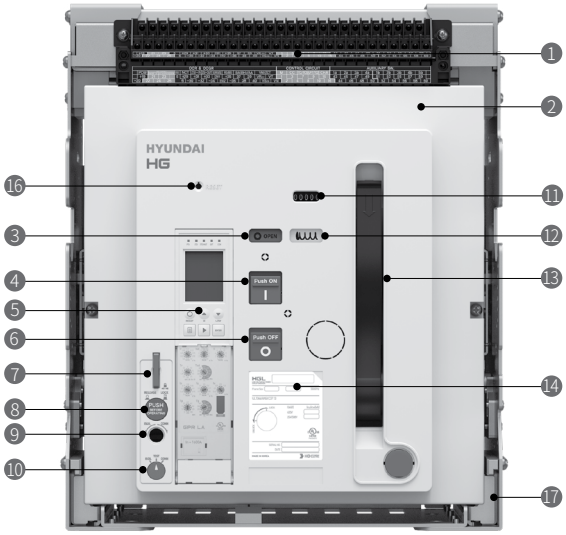


02 Structure

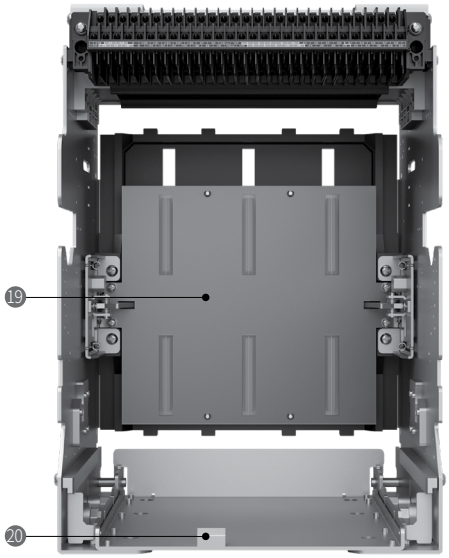
Fig. 1



Fixed Type (ACB Body)



Draw-In/Out Type (Including Cradle)



Cradle Front



Cradle Rear

❶ Control Circuit Terminal
❷ Front-Cover
❸ Close/Open Indicator
❹ Close Button
❺ Overcurrent Relay Device
❻ Open Button
❼ Position Padlock
❽ Position Lock Release Button
❾ Draw-In/Out Handle Insertion Hole
❿ Position Indicator
⓫ Counter
⓬ Charged/Discharged Indicator
⓭ Manual Charging Handle
⓮ Rating Nameplate
⓯ Terminal Busbar
⓰ OCR & Alarm S/W Reset Button
⓱ Draw-In/Out Guide Rail
⓲ Arc Shield
⓳ Safety Shutter
⓴ Auto Discharge Block

03 Operation

Air circuit breakers are available either manual charging type or motor charging type.

3.1 Manual Charging Type

In the manual charging type, both charging the closing springs and open-close control of the breaker should be done manually.
The breaker is able to close only when the closing spring is charged.

⚠ CAUTION

Do not force down the charging handle after completion of charging.
It may cause a malfunction.

1) Spring charging operation
Follow the procedure given below to charge the closing springs.
- Pump the charging handle (Fig. 2. ①) about five times.
- When closing springs are fully charged, a metallic click will be heard and no more pumping of the charging handle will be possible. Check that the spring charged indicator now shows「⏦」(Fig. 2. ②).

2) Closing operation
Check the instruction before closing the breaker.
- Closing spring should be charged.
- Position lock release button should be in original position (Fig. 3. ①).
- Specified voltage should be applied to the under-voltage trip device option.
  See the description of the under-voltage trip device for the procedure in detail.

Upon satisfactory confirmation of the above three items, press the closing button (Fig. 3. ②).
Then, closing springs are discharged and the breaker is closed.
The close/open indicator indicates「|」and spring charged indicator shows「⏦」(Fig. 3. ③).

3) Opening operation
Press the open button (Fig. 3. ④), close/open indicator indicates 「O」(Fig. 3. ⑤).

Fig. 2

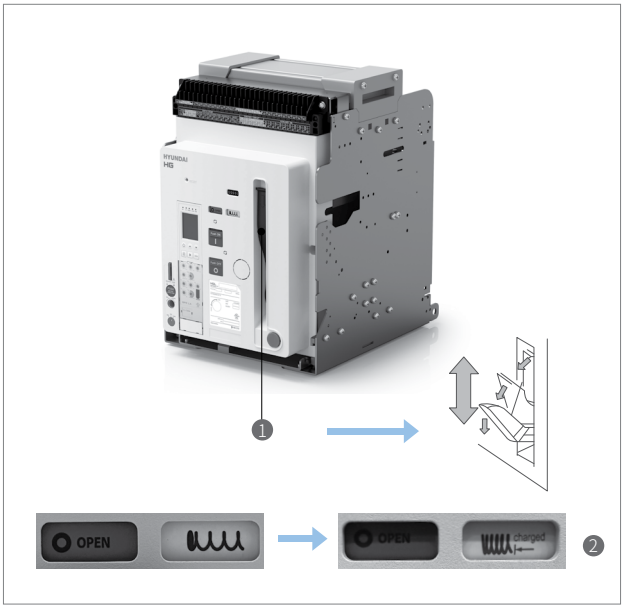
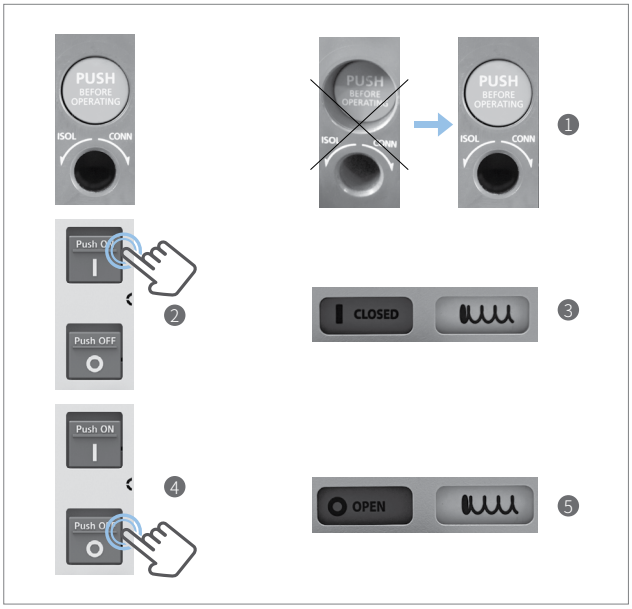


Fig. 3



3.2 Motor Charging Type

In this type, a motor-operated mechanism automatically charges the closing springs.
Means for remote electrical close-open control of the breaker are also fitted (see page 50-51 wiring circuit).
Manual operation is also possible (manual operation procedures are described in the 3.1 section manual charging type).

1) Spring charging operation

- Supply the specified control power voltage to the charging motor circuit.
- As soon as the closing springs are discharged, the charging motor is turned on to charge the closing springs.
- The charging motor is automatically stopped after the closing springs are fully charged.
  Then, charged/discharged indicator show charged sign,「⏦⊣」(Fig. 2. ②).
  Spring charging time is different according to control power and type of breaker.
  This is normally within a range of 3 to 10 seconds.

⚠ CAUTION

The permissible control voltage range for the charging motor is from 85 % to 110 % but 100 % of rated voltage is more recommended to be supplied.
Failure to follow this instruction will result damage in charging motor.

⚠ CAUTION

It is strongly recommended to supply the control power at the rated voltage.
Dielectric withstand test in motor, AC/DC 100-220 V : 1,500 V 1 min, DC 24, 48 V : 500 V 1 min.
The motor jack should be disconnected before dielectric withstand test.

2) Closing operation

Before closing the breaker, check the followings.

- Closing springs are charged.
- Position lock release button should be in original position.
- Under voltage trip device (UVT) is supplying rated voltage.

Upon satisfactory confirmation of the above items, press the PB (Close) button (page 50-51).
This energizes LRC (latch release coil, Fig. 9. ④) which in turn release the closing springs and close the breaker.
Close/Open indicator indicates「|」(Fig. 3. ③).
Charged/Discharged indicator indicates「⏦」(Fig. 3. ③).
When closing springs are discharged, the charging motor immediately turn to charge closing springs.

⚠ CAUTION

Even when no. 2) is not satisfied, the breaker will not be placed in a closed state.
Be sure to perform closing operation according no. 2).

3) Opening operation

The shunt trip device (SHT) or under voltage trip device (UVT) is used for remote electrical opening operation.
Press the open button (PB open, page 50-51). This opens the breaker via the SHT or UVT.

⚠ CAUTION

When close-open operations are repeated with the charging motor ON, limit the number of successive close-open cycles to 10 times. If the open-close cycle is repeated more than 10 times, allow a cooling period of at least 10 minutes between the 10th cycle and the 11th cycle.
Repeating open-close cycle more than 10 times consecutively may damage of charging motor.

04 Draw-out Mechanism

In order to test the breaker and exchange parts easily, the breaker body is drawn-in/out from draw-out cradle, the breaker can be fixed in one among three of draw-out cradle.
The breaker can be moved to TEST or ISOLATED position when the panel door is closed.

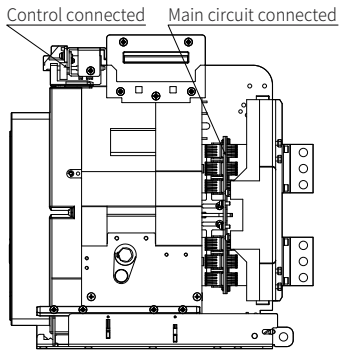
Caution of operation

- Open the breaker before working with the draw-out mechanism.
- Loosen the screws of the fixing blocks before drawing out the breaker body.
- Push the position lock release button (Fig. 1. ⑧), insert draw-in/out handle into the insertion hole (Fig. 1. ⑨).
- When you try to move the breaker body to the CONNECTED position, the operating force will increase.
  Torque of operating force is about 25 kgf.
- Operate draw-in/out handle totally in inserted state.
- Rotate draw-in/out handle until the handle is locked automatically at each ISOLATED, TEST, and CONNECTED position.
  At each locked position, stop turning the draw-out handle.
  Failure to follow this instruction can result in equiptment damage or malfunction.

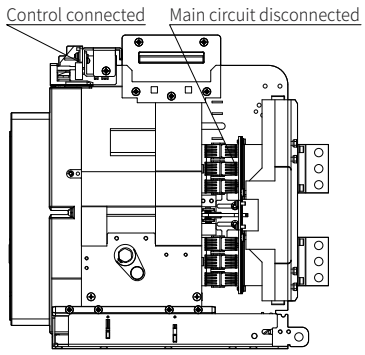
⚠ CAUTION

During the breaker body move from ISOLATED to CONNECTED position, or from CONNECTED to ISOLATED position, do not rotate the handle by releasing position locking.

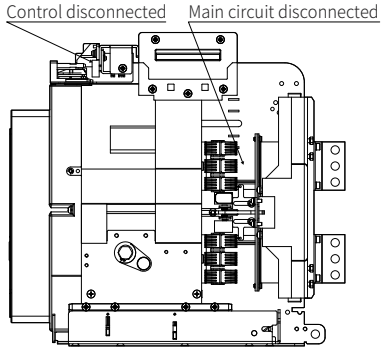
CONNECTED Position



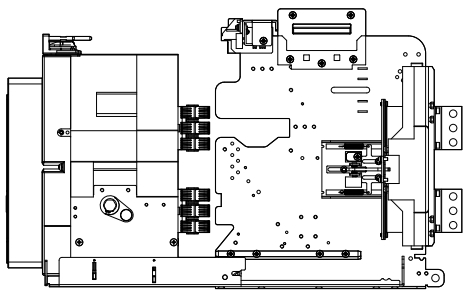
TEST Position



ISOLATED Position



REMOVED Position



4.1 Draw-out Operation

The draw-in/out handle is used to move the breaker body to one of the three positions (CONNECTED, TEST, ISOLATED).

⚠ CAUTION

When the fixing blocks (Fig. 4. ② option) are fitted, loosen the right and left screws of the fixing blocks before drawing-out operation.

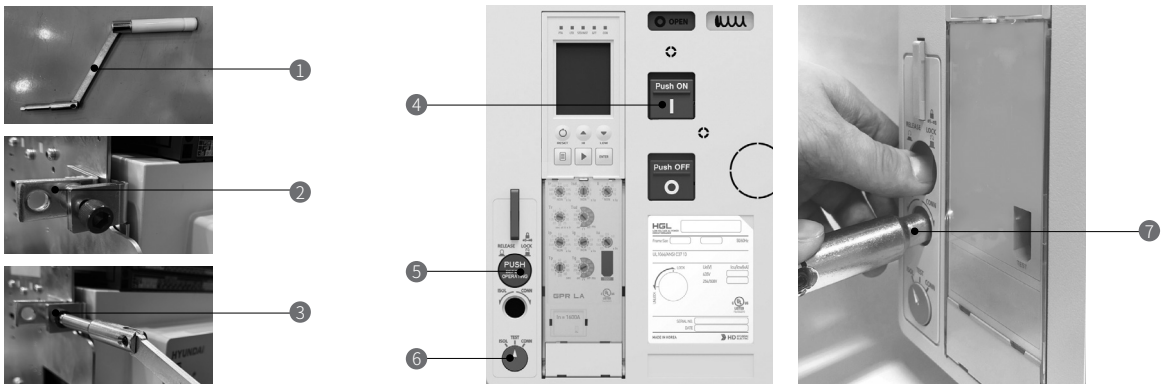
1) Moving from CONNECTED position to TEST position

- When the fixing blocks are fitted, loosen and free right and left screws. (Fig. 4. ③)
- Check that the breaker is open.
  If it is closed, push open button (Fig. 4. ④) to open breaker.
- When the position lock release button is completely pushed in, connect draw-in/out handle to shaft fully. (Fig. 4. ⑦)
  If it is not completely pushed in or not inserted, the body doesn't move and indicator doesn't rotate.
  The position indicator is rotated while the breaker body drawn out.

⚠ CAUTION

When the main circuit is disconnected as the breaker body is being drawn out, the breaker body will be slightly pushed forward by the spring action of the primary disconnect contacts with a "banging" sound.
While the loudness of this sound may vary, the sound itself is perfectly normal and does not affect breaker performance.

Fig. 4



⚠ CAUTION

When position lock release button (Fig. 4. ⑤) is pushed, the breaker will not close.
In order to test operation, must move to the right position (CONNECTED, TEST, ISOLATED), and position lock release button (Fig. 4. ⑤) should be pushed out.

⚠ DANGER

When breaker body is drawn out to the TEST position or ISOLATED position, a metallic sound "click" will be heard, and the draw-in/out handle operation is automatically locked.
At this time, do not attempt to compulsively rotate the handle.

04 Draw-out Mechanism

2) Moving from test position to Isolated position
In order to move from test to Isolated, push position lock release button (Fig. 4. ⑤) shortly after handle locking is released, and turn the draw-in/out handle in a counterclockwise direction.

⚠ CAUTION

If it is approaching ISOLATED position, draw-in/out handle (Fig. 4. ①) will be automatically locked.
Do not attempt to compulsively rotate the handle.

3) Removing body from Isolated position
For maintenance, inspection, or exchanging parts, the breaker in the isolated position should be drawn out from the draw-out cradle.
If the breaker is charged, the breaker should release the closing springs, by using the manual close and open button.

- To remove the ACB body from the cradle in the ISOLATED position, grasp both sides of the ACB body (Figure 5, ④) and pull it out completely.
- When the circuit breaker is fully withdrawn, slowly remove it from the rails on both sides of the Draw-Out Cradle (Figure 5, ①).

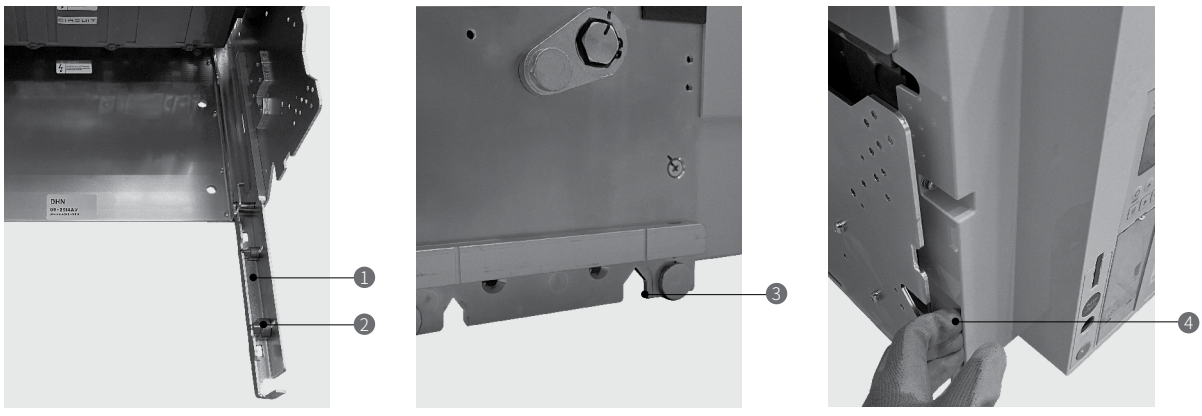
⚠ CAUTION

This operation should be done slowly.
Apply wire rope to the lifting lug and lift the body upward after releasing stoppers.

⚠ CAUTION

Don't leave the breaker in Removed position.
When the breaker is drawn out, the center of gravity is changed.
The draw-out cradle should be fixed in advance in the PNL.

Fig. 5



4.2 Draw-in Operation

In order to push back the breaker body to CONNECTED position, follow steps below.

- Check that the breaker is open. Also, check draw-in/out indicator shows isolated.
- Check charged/discharged indicator.
  Indicator shows discharged「」.
- Withdraw the rails on both sides of the Draw-Out Cradle (Figure 4, ①).
- Lift the circuit breaker using a dedicated lifter or by hanging it, and insert the circuit breaker into the Cradle Rail (Figure 4. ①), ensuring that the Side Guide of the circuit breaker body (Figure 4. ③) fits perfectly into the Guide of the rail (Figure 4. ②).
  Push the circuit breaker all the way in, but do not apply excessive force.

⚠ CAUTION

Never enter under the breaker.
The breaker may fall and cause serious injury.
Do not push hard the breaker.

- With position lock release button (Fig. 4. ⑤) pushed, connect drawing in/out handle to the shaft.
- Then, when position lock release button (Fig. 4. ⑤) is pushed, it maintains self holding and allows draw-in/out handle (Fig. 4. ①) operation.
- To move the breaker to Connected or Test position, rotate draw-in/out handle in a clockwise direction.
  With the breaker moving, if position indicator is accessing test position, position release button (Fig. 4. ⑤) is automatically pushed out and it locked draw-in/out handle.
  When the handle is locked, do not rotate excessively. This may cause damage to the breaker.
- In the test position, with the handle stick, PUSH position release button (Fig. 4. ⑤) again and release draw-in/out handles.
- With position lock release button pushed, it maintains self holding, rotate draw-in/out handle in a clockwise direction again.
- Main circuit connection starts just before CONNECTED position, draw-in/out handle operation will become heavy.
  But continue the rotating by adding force until, position indicator (Fig. 4. ⑥) of the breaker shows connected and, position release button (Fig. 4. ⑤) is automatically pushed out to lock the draw-in/out handle.
  Then, remove draw-in/out handle and fix it on the right side of draw-out cradle.
- In Connected position, do not operate draw-in/out handle in a clockwise direction. This may cause damage to the breaker.
- When the breaker fixing block is fitted, tighten left and right fixing screw by draw-in/out handle.
- In order to control the breaker electrically, you should connect all circuit according to the diagram on page 44-45.

05 Periodic Inspection and Parts Replacement

Period of periodic inspection

It is most appropriate that the user works out his or her own inspection plan for the breakers according to the switching frequency, the value of normal breaking and making currents, the magnitude of the fault current interrupted, service conditions, and environmental conditions.
It is recommended to perform a simplified inspection once every 6 months and a full inspection once every 12 months.
To perform an periodic inspection, draw out to ISOLATED position or move out of draw-out cradle.

Mechanical durability life of the breaker

The following table shows the mechanical durability of the breaker for Maximum Warranty Life.
When accumulated counting number of switching cycles excees those shown, we recommend you check the breaker.
For renewal or thorought inspection, please contact us.

Frame Size	Up to 2,500 AF	Between 3,200 AF and 4,000 AF	At 6,000 AF
Number of switching cycle	Total : 12,500 (Mechanical : 9,700, Electrical : 800)	Total : 4,000 (Mechanical : 3,200, Electrical : 800)	Total : 1,500 (Mechanical : 1,100, Electrical : 400)

* Electrical endurance tests were performed at the minimum rated current to reflect the most demanding arc interruption conditions.

Frequency of inspection

Frequency of inspection is considered with service condition and switching numbers, breaking/making current value, but it is recommended to perform a simplified inspection once every 6 months and a full inspection once every 12 months.

Inspection by switching numbers

Switching Conditions of ACB	Inspection Interval Based on Switching Cycles		
	Below 1,000 AF	1,250-2,500 AF	Above 3,200 AF
Switching operation in the state of nearly no-carrying current	1,000	1,000	1,000
Switching operation in the rated current region	500	500	100
Switching operation in overload region (about 2-3 times the rated current)	25	25	10
Switching operation in current interruption region	each time	each time	each time

⚠ CAUTION

Draw out the breaker to the ISOLATED position or remove the breaker body from the draw-out cradle for inspection or parts replacement purposes.
Make sure that residual heat of terminal should be cooled down before performing inspection work.
Unless it may cause burn.

5.1 Arc Chamber

Check each arc chamber during the periodic inspection and also, after a fault current take place.
A cracked arc chamber cover or de-ionized grid side plate, or a heavy, hard-to-remove molten contact or de-ionized pieces inside of the arc chamber, requires replacement of the arc chamber unit.

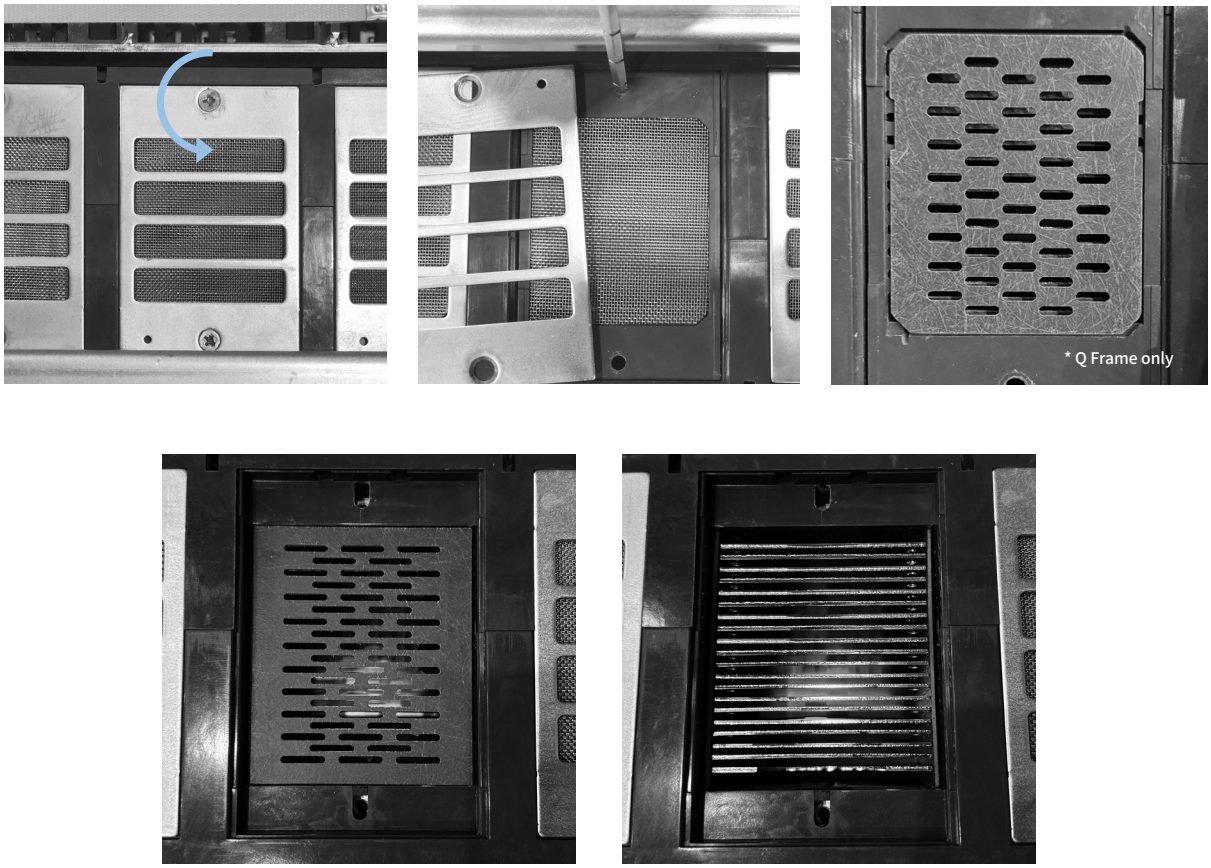
1) Periodic inspection

Inspection Item	Method / Criteria
Dirt, dust, foreign matter	Check visually. Inside must be clean, free of foreign matter and dust. Blow off foreign matter and dust with a jet of compressed air.
Cracks	Check visually. There should be no cracks or other damage.

2) Removal and mounting
Loosen the two-mounting screws on the arc chamber cover until they are free. Remove the arc chamber cover and nut.
To mount the arc chamber, set nut and arc chamber in position and tighten with two-mounting screws on the arc chamber cover.

To mount the arc chamber, check square nut, place the arc chamber in position and tighten fixing screw (dish M6).

Fig. 6

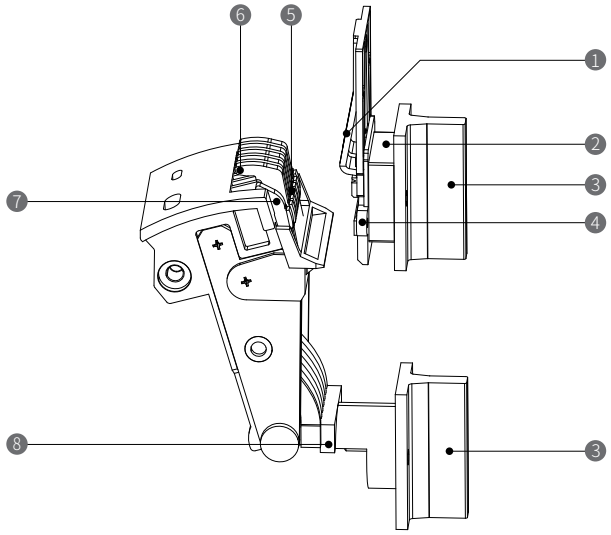


05 Periodic Inspection and Parts Replacement

5.2 Contact Unit

Contact unit are visible and accessible when the arc chamber is removed.
Check the contact unit during a periodic inspection and after a short-circuit fault current take place.

Fig. 7



1 Arc runner
2 Up terminal
3 In/out terminal body
4 Fix Contact Tip
5 Move Contact Tip
6 Arc move contact
7 Main move contact
8 Low terminal

1) Periodic inspection
(1) Arcing contact

Inspection Item	Method / Criteria
Contact tip surface	- Check visually. - Melting mark in the contact point tip surface is not a problem because it is a normal phenomenon by close-open arc. - Remove dirt, dust, grease, etc. - When the arcing contact is melted less than 33 % original thickness, it should be replaced. For replacements, please contact us.

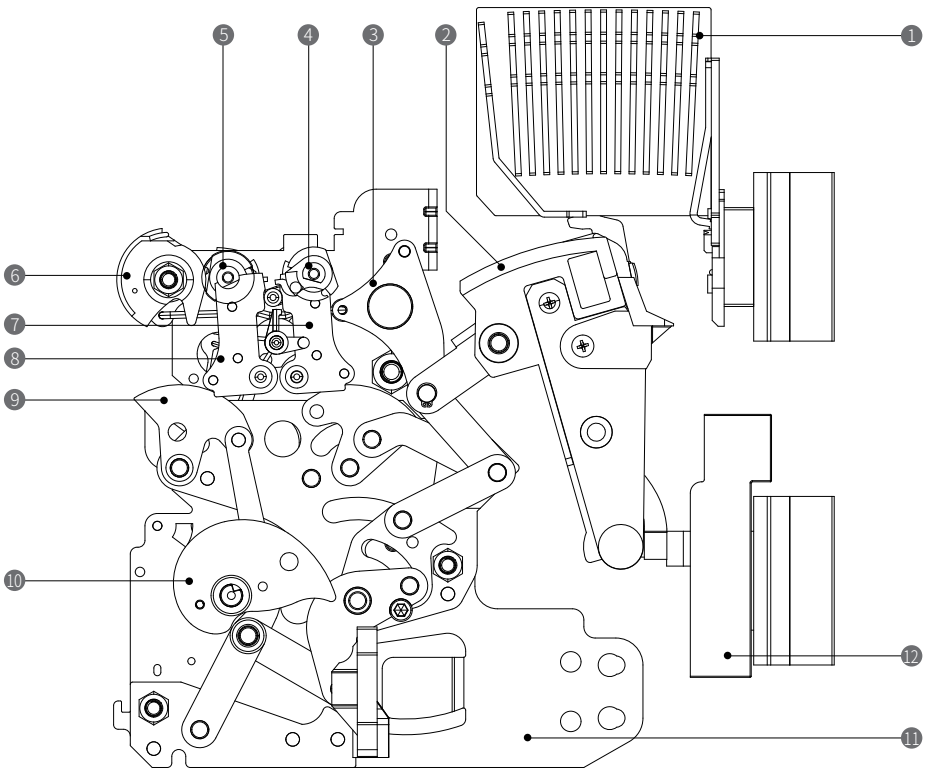
(2) Main contact
If contact surface is badly worn or become rough, you should clean the surface in case of periodic inspection.

⚠ CAUTION

When scrubbing contact trips' surface, pay attention to not let dust fall into the breaker mechanism.
After dressing, be sure to wipe and clean contact trips.

5.3 Operating Mechanism

Fig. 8



❶ Arc chamber	❺ Close lever	❾ CO latch
❷ Move contact fix base	❻ Charged/Discharged indicator	❿ Closing cam
❸ Crossbar unit	❼ Open latch	⓫ Mechanism frame
❹ Open lever	❽ Close latch	⓬ Current transformer

1) Periodic maintenance
Check the operating mechanism in detail as much as possible.
If there are parts that seem to be checked or hard to check, please contact us.

Inspection Item	Method / Criteria
Operation	Open and close breaker through manual control to check the mechanical parts for normal operation (every six months).
Lubrication	Add a small amount of grease to each of the pins, shafts and their bearings (every six months). Avoid excessive oiling to prevent the accumulation of dirt and dust.
Screws, bolts and springs	Check screws and bolts on each part for signs of loosening. Tighten them if loose. Check each spring for proper engagement and damage. Correct problem through repair or replacement (every six months).
Dirt and dust	Latching parts should be free of dirt and dust. Wipe them with a clean cloth (every six months).

05 Periodic Inspection and Parts Replacement

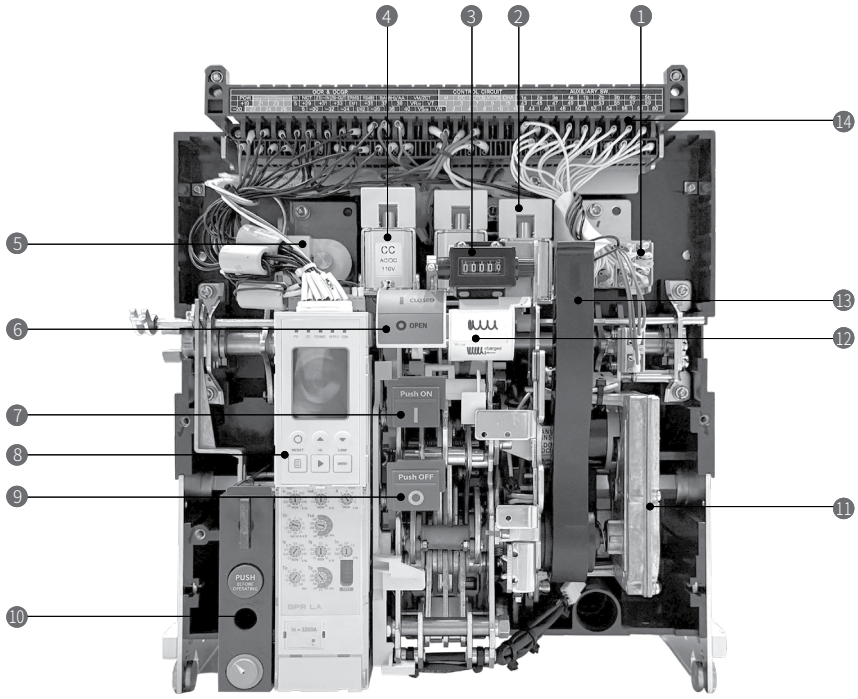
5.4 Internal Accessories

Loosen the two front cover mounting screws and open the front cover (Fig. 1. ②) to check internal accessories.

⚠ CAUTION

Do not place your fingers or a tool in the gap between close/open and charged/discharged indicator (Fig. 9. ⑥, ⑫) since this gap is closed when the breaker is closed.
Do not attempt to put hand or a tool into the breaker when closing springs are charged.
Be sure to discharge the closing springs prior to internal inspection.
Do not operate the on-off operation arbitrarity (qualified personnel only).

Fig. 9



❶ Auxiliary switch
❷ Shunt (SHT)
❸ Cycle counter
❹ LRC (latch release coil)
❺ MHT (magnetic hold trigger)
❻ Close/Open indicator
❼ Close button
❽ Over current relay (GPR)
❾ Open button
❿ Device of drawing in/out
⓫ Spring charging motor
⓬ Charged/Discharged indicator
⓭ Manual charging handle
⓮ Automatic connection control jack

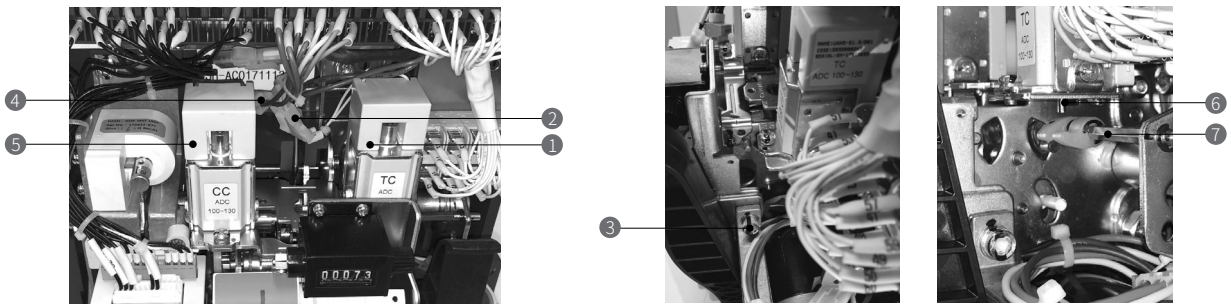
1) Shunt trip device (SHT)

Rated Voltage		Coil Resistence
AC/DC	110	40 Ω ± 5 %
	220	68 Ω ± 5 %
AC	380	250 Ω ± 5 %
	440	250 Ω ± 5 %
DC	24	7 Ω ± 5 %
	48	11.8 Ω ± 5 %

(1) Periodic inspection (Fig. 21. reference)

Inspection Items	Methods / Criteria / Dispositions
Operation	- Push the moving core of SHT with a pointed tool, such as a screwdriver trip, then, slowly release the core. The result is acceptable if the core returns lightly. - Push in the moving core slowly after closing the breaker. The result is acceptable if breaker trips open. - The result is acceptable, if the breaker trips open at less than 70 % of the rated voltage after closing the breaker.
Coil	- Measure the coil resistance with an ohmmeter. If it is much lower than the value shown in table or there is no continuity, replace the coil.
Terminals and mounting screws	- Check terminals and mounting screws. Tighten them if loose.

Fig. 10. Shunt trip equipment mounting



❶ Shunt coil
❷ Shunt terminal
❸ Shunt/LRC mounting support fixing bolt (R/H, M4)
❹ LRC terminal
❺ LRC coil
❻ Shunt moving core-trip rod assembly check
❼ Trip-rod

(2) Replacement of SHT (Fig. 10)

- Remove the two pin terminal from the control jack terminals (Fig. 10. ②).
- Remove the one support mount screws (Fig. 10. ③) and take out LRC-UVT-SHT support.
- Remove the coil mount screws (Fig. 13. ③).
- A new coil is properly assembled, and during attaching it. The trip-rod and shunt moving core position is shown as in Fig. 10. ⑥, ⑦.
- After checking the light movement of the moving core of new SHT, mount it in the breaker as it was.
- Connect the two pin terminal to the control jack terminals, and test the ACB and trip lever both electrically and mechanically for normal operation (number 7, 8).

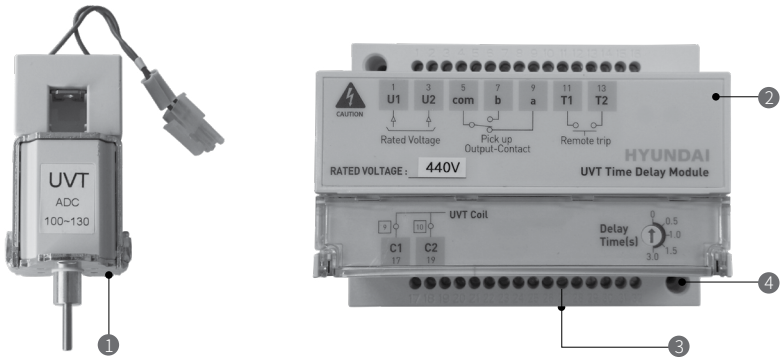
05 Periodic Inspection and Parts Replacement

2) Undervoltage trip device (UVT)

- UVT coil uses only DC rating.
- The under-voltage trip device includes instantaneous trip type and time delay trip type having 0.5 sec, 1 sec, 1.5 sec and 3 sec time delay characteristics.

(1) General view of undervoltage trip coil & controller (time delay trip type)

Fig. 11. UVT coil & UVT controller (only time delay type applied)



1. UVT coil
2. UVT controller
3. Wire terminal
4. UVT controller mounting hole

(2) Periodic inspection

Inspection Items	Methods / Criteria / Dispositions	
Operation	- The breaker can not be closed when the UVT is de-energized. Because of this, if the breaker can be closed at more than 85 % of the rated voltage, moving core becomes pick-up, the result is acceptable. - If the breaker can be tripped from 35 % to 70 % of the rated voltage after closing the breaker, the result is acceptable.	
Coil resistance	- Measure the coil resistance with an ohmmeter. If the measured resistance is much lower than the value shown in the right or there is no continuity, replace the UVT coil.	Coil resistance (Ω) 118 Ω ± 5 %
Terminals and mounting screws	- Check terminals and mounting screws for loosening. Retighten them if loose.	

(3) Replacement of UVT controller

When it is necessary to replace the UVT controller, replace the whole UVT controller unit.

- Remove the UVT wire at the wire terminal (Fig. 11. ①).
- Remove UVT mounting bolt (2-M6x10) (Fig. 11. ④).
- Take out the UVT controller.
- Install new UVT controller as it was on the breaker with UVT mounting bolt (Fig. 11. ④).
- Connect the wire terminals with fasten terminals at the same time.
- Test the UVT both electrically and mechanically.

⚠ CAUTION

When inspecting the performance, rated voltage should be supplied unless ACB will not operate.

⚠ CAUTION

Recheck the wire terminal before suppling power.
Incorrect wiring may cause coil burning.

3) Latch release coil (LRC)
Coil resistance is same with SHT.

(1) General view of latch release coil (LRC)

Fig. 12



❶ Terminal
❷ Moving core
❸ LRC mounting bolt tap (M5 tap. 1ea)

(2) Periodic inspection

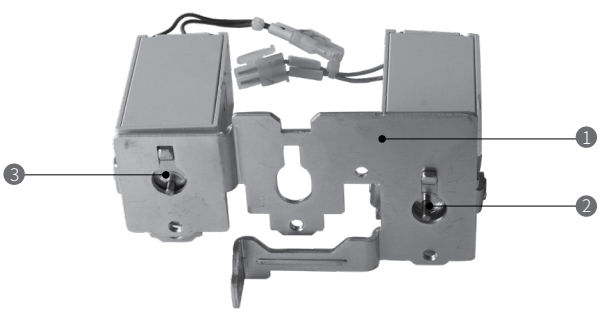
Inspection Items	Methods / Criteria / Dispositions
Operation	- Push the moving core with a pointed tool, such as a screwdriver tip, then, slowly release the core. The result is acceptable if the core returns lightly. - Charge the closing springs and push the moving core. The result is acceptable if the closing springs are discharged. - If the breaker can be closed at 35 % of the rated voltage, the result is acceptable.
Coil resistance	- Measure the coil resistance with an ohmmeter. If resistance was much lower than the value shown in table.1 (page 19) or there is no continuity, replace the CC.
Terminals and mounting screws	- Check terminals and mounting screws for loosening. Tighten them if they are loose.

(3) Replacement of LRC (unit replacement)
- Remove the pin terminal from the control jack terminals (Fig. 10. ④).
- Remove support mounting screw (Fig. 13. ②) M4, and take out the LRC-UVT-SHT support.
- Remove coil mounting screw (Fig. 13. ②).
- After checking the movement of the new LRC moving core, mount it in the breaker as it was.
- Connect the pin terminal to the control jack terminals (Fig. 10. ④).
- Test the device both electrically and mechanically for normal operation in the same manner described in periodic inspection table.

⚠ CAUTION

Take care when checking on/off operation by using mechanism manual button.
Fingers may be pinched during operation.

Fig. 13



❶ Shunt/CC mounting support
❷ LRC mounting (R/H M5)
❸ TRIP mounting (R/H M5)

05 Periodic Inspection and Parts Replacement

4) Auxiliary switch unit

(1) General view of auxiliary switch unit, connection circuit

Fig. 14. Aux, switch general view

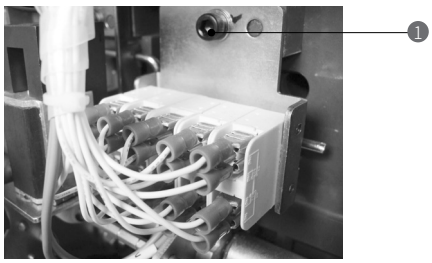
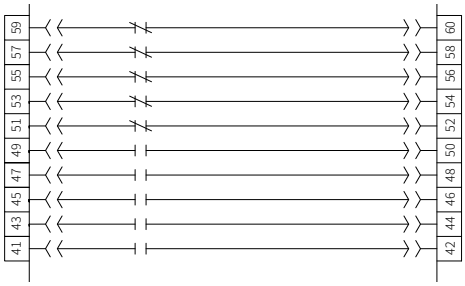


Fig. 15. Aux. switch connection circuit



(2) Periodic inspection

Inspection Items	Methods / Criteria / Dispositions
Operation	- Take the breaker body out of the cradle. - Connect the ohmmeter or alarm (buzzer) to each switch element. - Check that a-contact is ON and b-contact is OFF when breaker is CLOSED, and that a-contact is OFF and b-contact is ON when breaker is OPEN.
Contact surface	- If contact is excessively worn or rough, replace whole switch unit.
Terminals and mounting screws	- Check terminals and mounting screws. Tighten them if loose.

(3) Replacement of auxiliary switch unit (Fig. 17)

The auxiliary switch unit is an assembly of 10 circuits (5 a-contacts and 5 b-contacts).

Replace the whole switch unit even if there is a partial defect.

- Remove the auxiliary switch assembly bolt (1-M6 wrench bolt) (Fig. 14. ①).
- Change wire circuit.
- Tighten the auxiliary switch assembly bolt (1-M6 wrench bolt).
- Conduct periodic inspection mentioned in (2) above and make sure if the switch operates normally.

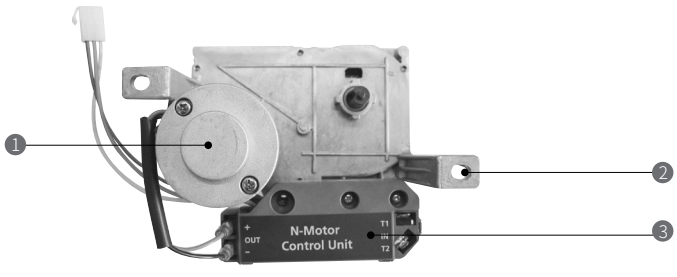
⚠ CAUTION

Replaced terminal must be properly connected before operating.

5) Motor unit

(1) Motor unit general view

Fig. 16



❶ Motor rotor
❷ Motor mounting hole
❸ Motor control unit

(2) Operation check

Open and close the breaker through electrical control to check the relay for normal operation.

Do this in the following procedure.

- After checking front-cover, with the push-button switch ON, close the breaker.
- Normal operation is when the breaker is closed, the motor is rotating and the charged/discharged indicator displays「MWW-|」.

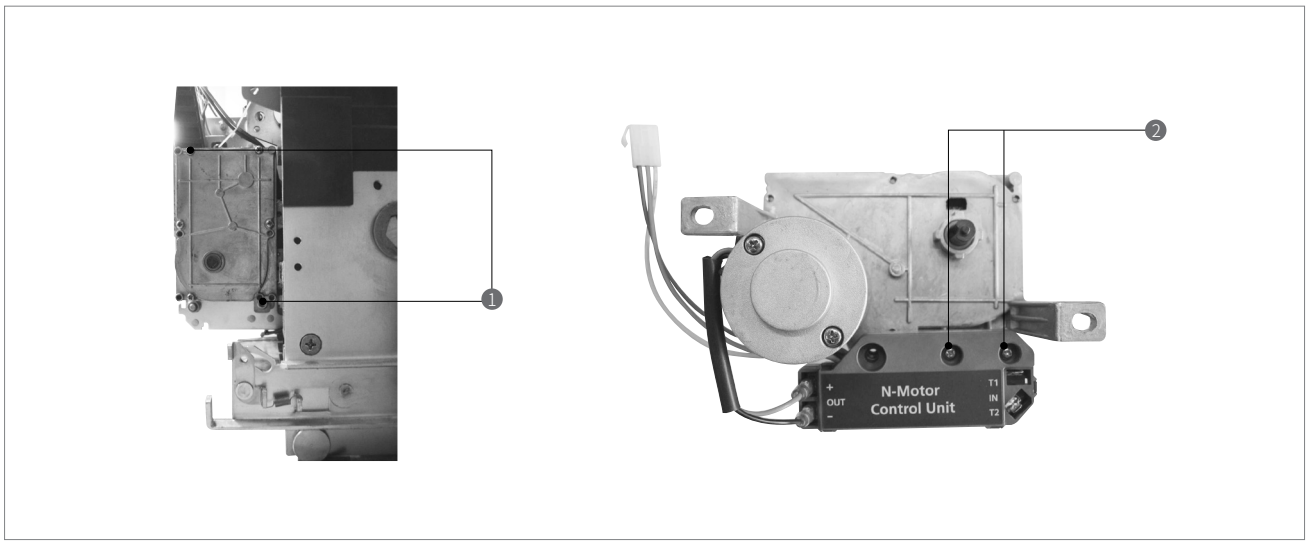
(3) Replacement of motor control unit

- Open the ACB.
- Draw-out the breaker body out of draw-out cradle.
- Remove front cover.
- Remove both motor cables (1, 2 terminal) and control jack terminal.
- Release motor fixing bolt (2-M6 wrench bolt, Fig. 17. ①).
- Remove motor from the breaker.
- Remove motor connection terminal (Fig. 17. ②).
- After checking new motor unit, replace, and follow steps 4-7 in reverse.
- Charging motor is originally placed position of the breaker.
- Attach front cover.
- Check the operation according to operation check (2).

⚠ CAUTION

Replaced line terminal must be properly connected before operation.

Fig. 17



6) Direction of ACB position padlock device

- Purpose: Position padlock device is designed for locking ACB body at three positions (ISOLATED, TEST, CONNECTED) by a shackle.
- Usage: At any position, pull out position padlock plate and attach the shackle (user supplied).

⚠ DANGER

Do not draw out ACB when it is in the service states.
Since main circuit is live, draw-out operation will cause fatal accident.
Check the breaker is opened before operating draw-in/out.

05 Periodic Inspection and Parts Replacement

5.5 Periodic Inspection

1) Special : Fault current cut-off or open and close more than 1,000 times/year
2) Usually : Within 2 years of installation
3) Thorough : More than 3 years after installation

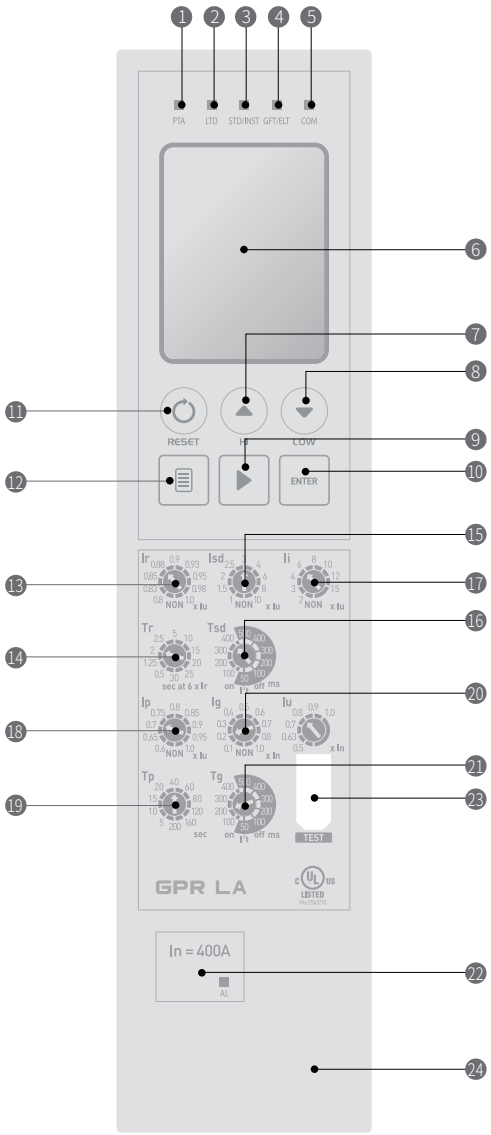
Inspection Part	Inspection Items	Inspection Cycle			Inspection Details
		Special	Usually	Thorough	
ACB Body exterior	Appearance inspection		1 year		1) Procedures : - Check for insulation mold, control jack, front cover, OCR damage or crack - Check for damage to optional devices such as button lock, position switch, key lock, etc - Check that the body is contaminated with dust and foreign substances - Check ACB body terminal discoloration, contamination, and bolt fastening 2) Judgment : No damage, cracks, dust or foreign substance **⚠ CAUTION** **Contamination with dust or foreign substances may cause faulty operation or insulation performance deterioration.**
ACB Body terminal part			1 year		1) Procedures : - Check the level of discoloration of the main body terminal and contamination by foreign substances - Check bolt fastening status 2) Judgment : No abnormality (M8 bolt fastening torque: 135kg·cm) **⚠ WARNING** **In case of contamination or bolt loosening, Overheating may be occurred due to increased resistance in the main circuit.**
Arc Chamber		After cutting off fault current	1 year		1) Procedure : - Check for DI GRID unit damage or cracks(Including inspection for dust and conductive foreign substances) 2) Judgment : No damage or contamination
ACB Body contact unit		After cutting off fault current	1 year		1) Procedure : Check for Arc move contact, main move contact damage or cracks (including checking for dust and foreign substances) 2) Judgment : No damage or contamination **⚠ WARNING** **In case of contact contamination, Overheating may be occurred due to increased resistance in the main circuit(If contacts are damaged, parts or products need to be replaced).**
Mechanism		1,000 times	1 year		1) Procedures : - Check internal parts for damage and cracks - Check operation part for sticking of dust and lubrication grease - Check the mechanism manual and electric ON/OFF and charge functions (5 times each) - Check mechanism bolt fastening status 2) Judgment : No abnormality (M8 bolt fastening torque: 135kg·cm) **⚠ CAUTION** **Defective operation may occur when bolts are loosened and lubricated grease is hardened.**
Cradle			1 year		1) Procedures : - Check for insulation mold, control jack, left/right side plate parts damaged, cracks - Check safety shutter operation and check for damage - Isol contact to check for discoloration, grease sticking and contamination with foreign substances 2) Judgment : No abnormality **⚠ CAUTION** **In case of Isol contact contamination, Overheating may be occurred due to increased resistance.**
Closing Coil Trip Coil UVT Coil		1,000 times	1 year		1) Procedures : - Check cable sheath damage, cable breakage and connector connection - Check if the coil internal operation rod works smoothly up/down 2) Judgment : No abnormality
OCR	OCR operation test	After cutting off fault current	1 year		1) Procedures : - Check the OCR Display for any abnormal load current and measurements - Conduct a test of the OCR Tester kit long-term, short-term, instantaneous, and ground fault functions to check the operation - Check output contact points for each function when checking OCR operation (※ Refer to the catalog and user manual for operation time standard) 2) Judgment : The measured tolerance is less than 5% and the OCR function and contact output are not abnormal
CRADLE External Terminals	Load inspection	After cutting off fault current	1 year		1) Procedure : - Check Cradle external terminal temperature with thermal imaging camera and check if there is any abnormality 2) Judgment : Measurement temperature must be 70K or less (K = Measurement temperature - External temperature) **⚠ WARNING** **If it exceeds 70K, overheating is suspected, and it is necessary to contact HD HYUNDAI ELECTRIC for inspection.**

<table>
<tr><th rowspan="2">Inspection Part</th><th rowspan="2">Inspection Items</th><th colspan="3">Inspection Cycle</th><th rowspan="2">Inspection Details</th></tr>
<tr><th>Special</th><th>Usually</th><th>Thorough</th></tr>
<tr><td>Closing Coil
Trip Coil</td><td rowspan="4">Test of operation characteristics</td><td>1,000 times</td><td></td><td>3 year</td><td>1) Procedures :
- Check closing coil operation within the control voltage (0.85 - 1.1 Vn) range
- Check trip coil operation within control voltage (0.7 - 1.1 Vn)
2) Judgment : Nothing wrong</td></tr>
<tr><td>UVT Coil
UVT Controller</td><td>1,000 times</td><td></td><td>3 year</td><td>1) Procedures :
- Check UVT Coil operation at 0.85Vn or higher by stepping up the control voltage
- Check UVT coil operation at 0.35Vn or less by writing down the control voltage
- Check the UVT controller delay set operation time below the control voltage of 0.35Vn (0.5, 1, 1.5, 3 sec)
2) Judgment : Nothing wrong</td></tr>
<tr><td>Motor</td><td>1,000 times</td><td></td><td>3 year</td><td>1) Procedures :
- Check motor operation within the control voltage (0.85 - 1.1 Vn) range
- Check mechanism handle limit sw contact before/after motor charging
2) Judgment : Nothing wrong</td></tr>
<tr><td>Auxiliary switch</td><td></td><td></td><td>3 year</td><td>1) Procedures :
- Check auxiliary contact "a" contact status after ACB Close
- Check auxiliary contact "b" contact status after ACB Open
2) Judgment : No abnormal contact output</td></tr>
<tr><td>Main circuit of the ACB body</td><td>Main circuit contact resistance test</td><td>After cutting off fault current</td><td></td><td>3 year</td><td>1) Procedure :
- Measure the main circuit contact resistance for each phase and check whether there is an abnormality (based on DC100A)
2) Judgment : Below the standard
⚠ WARNING
If the main circuit resistance is higher than the standard value, Overheating may be occurred.
<table>
<tr><th colspan="3">HGL TYPE</th></tr>
<tr><th colspan="2" rowspan="2">TYPE</th><th>RESISTANCE(μΩ)</th></tr>
<tr><th>ACB BODY (Draw-out TYPE)</th></tr>
<tr><td>A FRAME</td><td>HGL08~16</td><td>13</td></tr>
<tr><td rowspan="2">B, Q FRAME</td><td>HGL08~25</td><td>12</td></tr>
<tr><td>HGL32</td><td>12</td></tr>
<tr><td>C FRAME</td><td>HGL40</td><td>20</td></tr>
<tr><td>D FRMAE</td><td>HGL60</td><td>13</td></tr>
</table></td></tr>
<tr><td>Main circuit of the ACB body control circuit</td><td>Insulation resistance test</td><td>After cutting off fault current</td><td></td><td>3 year</td><td>1) Procedures :
- Conduct an insulation resistance test of the main circuit and control circuit to check if there are any abnormalities (applies 500V Megger)
2) Judgment : Main circuit 1000MΩ or more, control circuit: 300MΩ or more</td></tr>
<tr><td>Main circuit of the ACB body control circuit</td><td>Withstand voltage test</td><td>After cutting off fault current</td><td></td><td>3 year</td><td>1) Procedures :
- Check the withstand voltage performance of the main and control circuits
*Main circuit: AC3500V for 1 minute, Control circuit: AC1500V for 1 minute
(OCR is removed during test)
2) Judgment : Nothing wrong
⚠ CAUTION
If there is any abnormality in withstand voltage performance, please contact HD HYUNDAI ELECTRIC for inspection.</td></tr>
</table>

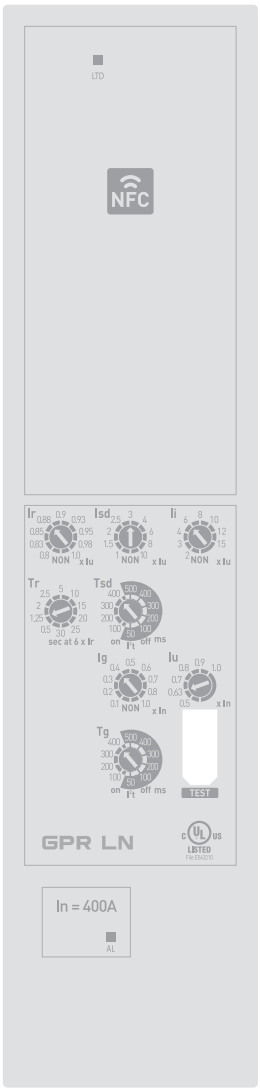
06 Protection Relay - GPR

Fig. 18

GPR-LA



GPR-LN



1. PTA signal LED
2. LTD signal LED
3. STD/INST signal LED
4. GFT/ELT signal LED
5. Com. signal LED
6. LCD
7. STD/INST test button
8. LTD test switch
9. Movement button
10. ENTER button
11. RESET button
12. LIST button
13. LTD pick up setting switch
14. LTD time setting switch
15. STD pick up setting switch
16. STD time setting switch
17. INST pick up setting switch
18. PTA pick up setting switch
19. PTA time setting switch
20. GFT/ELT pick up setting switch
21. GFT/ELT time setting switch
22. Rating plug
23. Temporary test
24. Battery

Enhancement of Over Current Relay (OCR) Functions Over current relay (OCR) built in the HG Series air circuit breaker has reinforced power monitoring functions such as temperature monitoring, fault recording other than the basic protection function, ultimately enabling stable power supply.

Item		General Feeder			
Model Name		N Type	A Type		P Type
		GPR-LN	GPR-LA	GPR-LAG	GPR-LP
Externals					
Frequency	50Hz	50	51	52	54
	60Hz	60	61	62	64
Main Function		- L/S/I/G - Thermal - Self-Power - Fail Safe - Fault Recording 10ea (Check Via NFC) - Last Fault's Waveform - Recording (4 Cycles, Check Via NFC)	- L/S/I/G - Thermal - Self-Power - Fail Safe - Communication (Modbus) - External Power - ZSI - Remote Reset Function - Individual Continuous Contact : LTD, STD/INST, GFT, PTA - 256ea Fault Recording - Last Fault's Waveform Recording (4 Cycles, Check Via Communication) - 200ea Event Recording (Check Via Communication) - ERMS	- L/S/I Uses Dedicated Outer CT Ground if Outer CT Ground (More than 30A) > 5 A at Secondary) - Thermal - Self-Power - Fail Safe - Communication (Modbus) - External Power - ZSI - Remote Reset Function - Individual Continuous Contact : LTD, STD/INST, ELT, PTA) - 256ea Fault Recording - Last Fault's Waveform - Recording (4 Cycles, Check Via Communication) - 200ea Event Recording (Check Via Communication) - ERMS	- L/S/I/G - Thermal - Self-Power - Fail Safe - Communication (Modbus) - External Power - ZSI - Remote Reset Function - Individual Continuous - Contact : LTD, STD/INST, GFT, PTA - 256 ea Fault Recording. - Last Fault's Waveform Recording (4 Cycles, Check Via Communication) - 200ea Event Recording (Check Via Communication) - OVR / UVR - unbalance Type (Voltage / Current) - Reverse Power - Voltage/Current - Power (P, Q, S), Power Factor - Energy - Demand - ERMS

06 Protection Relay - GPR

6.1 Rated Current

Type	Rated Current [I_n]
GPR-L Type	- Uninterrupted current [Iu] can be adjusted to 50 %, 63 %, 70 %, 80 %, 90 %, and 100%of the rated current [In]. - Rated current [In} is determined by rating plug. Rating plug is replaceable unit thatcan be adjusted to 50% ~ 100% of circuit breaker AF.

Rating plug	AF						
Rating	800AF	1600AF	2000AF	2500AF	3200AF	4000AF	6000AF
400A							
600A							
630A							
800A							
1000A							
1200A							
1250A							
1600A							
2000A							
2500A							
3000A							
3200A							
3600A							
4000A							
5000A							
6000A							

6.2 Protection

Long time												
Pick up between 1.05 ~ 1.20 Ir	Iu = In x		0.5	0.63	0.7	0.8	0.9	1				
	Ir = Iu x		0.8	0.83	0.85	0.88	0.9	0.93	0.95	0.98	1	non
Time delay (sec) Tolerance Pick Largest value between ±15% (Ir < 6In),±20% (Ir ≥ 6In), or ±40ms	tr @ (1.5×Ir)		10.4	26.1	41.7	52	104	208	312	417	521	626
	tr @ (6.0×Ir)		0.5	1.25	2	2.5	5	10	15	20	25	30
	tr @ (7.2×Ir)		0.35	0.86	1.38	1.73	3.45	6.9	10.4	13.8	17.3	20.7
Short time												
Threshold accuracy: ±10%	Isd = Iu x		1	1.5	2	2.5	3	4	6	8	10	non
Time delay (sec) @ 10xIu Accuracy: ±15% or ±40ms I2t on: ±20% or 40ms	tsd	I^2t off	0.05	0.1	0.2	0.3	0.4	0.5				
		I^2t on		0.1	0.2	0.3	0.4					
	I2t off	min. trip time	0.02	0.8	0.16	0.26	0.36	0.46				
		max. trip time	0.08	0.14	0.24	0.34	0.44	0.54				
Instantaneous												
Threshold (A) Accuracy: ±10%	Ii = Iu x		2	3	4	6	8	10	12	15	non	
Trip time			under 50ms									
Ground fault												
Threshold (A) Accuracy: ±15%(Ig > 0.4In) ±20%(Ig ≤ 0.4In)	Ig = In x * Ig cannot adjust over 1200A		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	1	non
Time delay (sec) @ 1 x In Accuracy: ±20% or 40ms	tg	I^2t off	0.05	0.1	0.2	0.3	0.4	0.5				
		I^2t on		0.1	0.2	0.3	0.4					
	I^2t off	min. trip time	0.02	0.8	0.16	0.26	0.36	0.46				
		max. trip time	0.08	0.14	0.24	0.34	0.44	0.54				
Pre-Alarm												
Threshold Note) Ig cannot adjust over 1200A.	Ip = Iu x		0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95	1	non
Time delay (sec) Accuracy: ±20%	tp		5	10	15	20	40	60	80	120	160	
Earth leakage												
Threshold	I Δn		0.5	0.8	1	2	3	5	Non			
Time delay (sec) Accuracy: ± 15 % or 40ms	Δt	Alarm Time	0.14	0.23	0.35	0.8	0.95					
		Trip Time	0.06	0.14	0.23	0.35	0.8					

06 Protection Relay - GPR

Protection		Threshold			Time delay		
		Range	Step	Accuracy	Range	Step	Accuracy
Under voltage	Pickup	125V ~ OVR	1V	±10%	1.2s ~ 5s	0.1s	±10%
	Dropout	Pickup ~ Umax	1V	±10%	1.2s ~ 36s	0.1s	±10%
Over voltage OVR	Pickup	UVR ~ 1200V	1V	±10%	1.2s ~ 5s	0.1s	±10%
	Dropout	125V ~ OVR	1V	±10%	1.2s ~ 36s	0.1s	±10%
Voltage unbalance	Pickup	2% ~ 30%	1%	±15%	1s ~ 40S	1s	±10%
	Dropout	2% ~ pickup	1%	±15%	10 ~ 360s	1s	±10%
Current unbalance	Pickup	5 ~ 60%	1%	±15%	1 ~ 40s	1s	±10%
	Dropout	5 ~ pickup	1%	±15%	10 ~ 360s	1s	±10%
Reverse power (Total active power)	Pickup	5 ~ 500 kW	1kW	±10% (> 0.2In) ±20% (≤ 0.2In)	0.2s ~ 20s	0.1s	+20% or 200ms
	Dropout	5kW ~ Pickup	1kW	±10% (> 0.2In) ±20% (≤ 0.2In)	1s ~ 360s	0.1s	+20% or 200ms

Note) Voltage protection is based on line-to-line voltage

6.3 Operation Indication Functions

The operation indication function includes contact for tripped indication (whole indication).

1) The contact for tripped indication (whole indication)

- This contact output is one contact operating even when of LTD, STD, INST, or GFT operates.

Rating of contact

Rating	Nominal Switching Capacity (Resistive Load)	5A 277VAC
	Max. Switching Power (Resistive Load)	1.385VA
	Max. Switching Voltage	277VAC
	Max. Switching Current	5A
	Max. Switching Capacity (Reference Value)	100mA 5VDC

2) LED light and contact for tripped indication (type GPR-LA, LAG, LP)

- Control power is required for tripped indication. Supply the following power between terminals ⑲ and ⑳ of control circuit disconnecting device (Fig. 1. ⑫).
  - AC 100-125 V / 10 VA or AC 200-250 V / 10 VA
  - DC 100-125 V / 10 W or DC 200-250 V / 10 W or DC 240 V / 10 W
- When LTD, STDTD, STD or GFT operates, the LED lights for tripped indication (Fig. 18. ②, ③, ④) light up individually.
- At the same time, the contact signal ON is outputted individually between terminals ㉒ - ㉑ of control circuit disconnecting device (Fig. 1. ①).

Operation signal contact terminal

㉒-㉑ : LTD contact terminal	㉒-㉔ : STD/INST contact terminal
㉒-㉓ : PTA contact terminal	㉒-㉕ : GFR/ELT contact terminal

- The LTD, STD and GFT tripped indications remain ON until the reset button (Fig. 18. 1⑪) is pressed.
  Pressing the reset button for more than one second resets the ON state.
- The PTA pickup indication LED light (Fig. 18. ①) flickers at more than pickup current value [I_P].
  At the same time, the contact signal ON is outputted between terminals ㉒-㉓ of control circuit.
- The operation indication is automatically reset if the current of less than [I_P] is reached.

※ Push the reset button (Fig. 18. ⑪).
If the abnormality is temporary, such as noise, the LED will turn off. At that time, the control function returns to normal.
If the LED does not go off after pushing the reset button, some abnormality may occur. In this case, immediately contact to HE.

- contact specification for tripped indication

Rating	Nominal Switching Capacity (Resistive Load)	5A 277VAC
	Max. Switching Power (Resistive Load)	1.385VA
	Max. Switching Voltage	277VAC
	Max. Switching Current	5A
	Max. Switching Capacity (Reference Value)	100mA 5VDC

6.4 Field Test

1) Function check method

- A simple check of operation for type GPR protection relay can be made with a HI/LOW button.

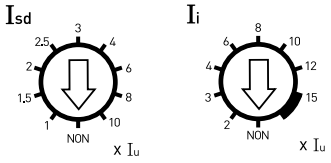
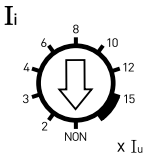
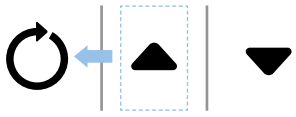
⚠ CAUTION

For function check of each protective function, draw out the breaker to the ISOLATED position, or take it out from the draw-out cradle.
If making function check in the TEST position, confirm that there is no influence on the sequence.
In state with load current, Field Test function is automatically off.

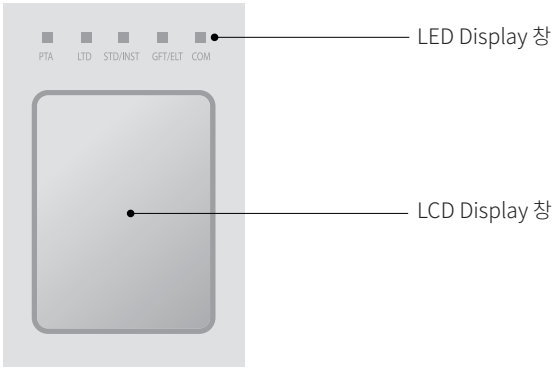
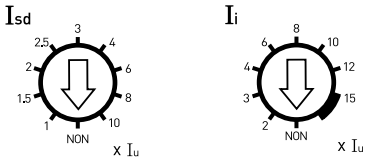
- Function check of the ground fault trip function is not possible (for ground fault trip function, portable UTK can be possible).
- Control power supply: Use marking plate of type GPR multifunction protective device if test in the test position checks function, check whether control power supply ⑲ to ⑳ of control circuit short device (Fig. 1. ⑫).

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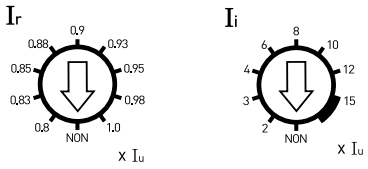
3) Protective function check

Protective Function	Type GPR-L (for general feeder circuit)
LTD	1. Close the breaker. 2. In case of GPR-L model, when short time delay and instantaneous time setting current value is lower than rated current, each dial should be in "NON" position using a flat-head screwdriver.  3. Push the "LOW" LOW Button (Fig. 18. ⑧) (current flow of 6 times the LTD [Ir] in type GPR-L and at the same time, measure the tripping time with a stopwatch. Also, hold the LOW Button until the breaker is tripped.  4. After tripping, release the LOW Button. After tripping, reset the OCR by pushing the RESET button (Fig 18. ⑪) 5. Read the tripped time. in case of type GPR-L, by flowing current (6 times of [I_r]), If it operates in allowable error of Tr setting time, it is normal. 6. After the LTD test, short time delay and instantaneous time setting should be reset to original setting value by flat head screwdriver.
STD	1. Close the breaker. 2. Instantaneous time trip setting current dial should set to NON position using a flat-head screw-driver.  3. Push the HIGH Button (Fig. 18. ⑦) to simulation currents greater 15 times the rated current [I_u]. If the breaker is tripped with Tsd setting time, it is normal.  4. After tripping, release the HIGH Button. Push the RESET button to reset fault information. Return the STD pickup current setting dial to the original set value with a flat-head screwdriver.
INST	1. Close the breaker. 2. Push the HIGH button (Fig. 18. ⑦) to simulation currents of more than 15 times the base current [I_u]. 3. If the breaker is tripped momentarily, it is normal. 4. After tripping, release the OCR by pushing the RESET button (Fig 18. ⑪)

6.5 Characteristic Check

Protective Function		Type GPR-L (for general feeder circuit)
LTD	Pick-up current	1. Close the breaker. 2. Input the current 1.2 times of the [Ir] setting current. LTD trip alarm LED (Fig. 18.②) is flickering. The range of Pick-up allowable error is ±10% 3. After checking for LED's Flicker, off the test current.  ※ LCD display is not available for type GPR-LN.
	Trip time	1. Close the breaker. 2. In case of type GPR-L, when short time and instantaneous time setting current value are lower than 6 times of rated current, each dial using a flat-bladed screwdriver should set NON position.  3. Type GPR-L flow 6 times current of [Ir] setting value. And at the same time, measure trip deadline using the stopwatch or portable GPR checker. 4. The breaker is tripped in a range of ±20 %, it is normal. 5. After checking trip operation, reset the OCR. 6. In case of operation of [2], short time delay and instantaneous time setting dial should be reset original setting value with a flat-head screwdriver.

06 Protection Relay - GPR

<table>
<tr><th colspan="2">Protective Function</th><th>Type GPR-L (for general feeder circuit)</th></tr>
<tr><td rowspan="2">Short time delay trip characteristic check</td><td>Trip setting current</td><td>1. Close the breaker.
2. Set long time and instantaneous time trip setting current dial to non using a flat-head screw-driver.
 
</td></tr>
<tr><td>Trip time</td><td>1. Close the breaker.
2. Set the test equipment output to 1.2 times the [Isd] setting current.
3. Apply test current. At the same time, start the trip delay measurement with a stopwatch.
4. If the breaker is tripped, reduce the test current to zero.
5. Read the trip time. If the trip time is within the range of resettable time (ms) and maximum total clearing time (ms) shown in the table below, it is normal.
6. Return the LTD/INST pickup current setting dial to the original set value with a flat-head screwdriver.

<table>
<tr><td rowspan="2">tsd</td><td>I^2t off</td><td>0.05</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td></tr>
<tr><td>I^2t on</td><td></td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td></td></tr>
<tr><td rowspan="2">I^2t off</td><td>min. trip time</td><td>0.02</td><td>0.8</td><td>0.16</td><td>0.26</td><td>0.36</td><td>0.46</td></tr>
<tr><td>max. trip time</td><td>0.08</td><td>0.14</td><td>0.24</td><td>0.34</td><td>0.44</td><td>0.54</td></tr>
</table></td></tr>
<tr><td>INST</td><td>Trip setting current</td><td>1. Close the breaker.
2. setting current and trips at +20 % [Ii] setting current, it is normal.</td></tr>
</table>

<table>
<tr><th colspan="2">Protective Function</th><th>Type GPR-L (for general feeder circuit)</th></tr>
<tr><td rowspan="2">Pre trip alarm</td><td>Pick-up current</td><td>1. Close the breaker.
2. Pre-alarm indicator LED(Fig. 18. ①) flickers when input the 100% current of the [I_p] setting value.
3. Reduce the test current to zero.
  </td></tr>
<tr><td>Trip time</td><td>1. Supply the specified control power.
2. Apply the test current of 1.1 times of the set value [I_p] current.
At the same time, start the time measurement.
- If the time at which an alarm signals is outputted across COM-PTA terminals is whin the range 20% of the set value [T_p] second, it is normal.
- Reduce of the test current to zero.
<table><tr><td>tp</td><td>5</td><td>10</td><td>15</td><td>20</td><td>40</td><td>60</td><td>80</td><td>120</td><td>160</td></tr></table></td></tr>
<tr><td rowspan="2">GFT</td><td>Trip setting current</td><td>1. Close the breaker.
2. Apply the test current of 1.1 times of setting value [Ig] current. If the breaker is tripped, it is normal.
3. Reduce the test current to zero.

Ig cannot adjust over 1200A</td></tr>
<tr><td>Trip time</td><td>1. Close the breaker.
2. Set the test equipment output to 120 % of the set value [In] current.
3. Apply the test current. At the same time, start the trip delay measurement with a stopwatch.
4. If the breaker is tripped, reduce the test equipment output to zero.
5. Read the trip time. If the trip time value is within the range of resettable time (ms) and maximum total clearing time (ms) shown in the table below, it is normal
<table><tr><td rowspan="2">tg</td><td>I^2t off</td><td>0.05</td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td></tr><tr><td>I^2t on</td><td></td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td></td></tr><tr><td rowspan="2">I^2t off</td><td>min. trip time</td><td>0.02</td><td>0.8</td><td>0.16</td><td>0.26</td><td>0.36</td><td>0.46</td></tr><tr><td>max. trip time</td><td>0.08</td><td>0.14</td><td>0.24</td><td>0.34</td><td>0.44</td><td>0.54</td></tr></table></td></tr>
<tr><td rowspan="3">Neutral protection</td><td>NON</td><td>1. Set when switchboard system does not needs neutral pole protection.
2. It is only possible with control power supply type of OCR.
3. It is impossible with GPR LN type.</td></tr>
<tr><td>50n (@0.5 I_n)</td><td>1. It operates at 50 % of In value.
2. LTD current of neutral pole is same as 50 % of Ir setting value.
3. STD current of neutral pole is same as 50 % of Isd setting value.
4. INST operates at 100 % of Ii setting value.
5. It is impossible with GPR-LN.</td></tr>
<tr><td>100n (@ I_n)</td><td>1. It operates at 100 % of In value.
2. LTD current of neutral pole is same as 100 % of Ir setting value.
3. STD current of neutral pole is same as 100 % of Isd setting value.
4. INST operates at 100 % of Ii setting value.
5. The GPR LN is set to 100% only.</td></tr>
</table>

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ERMS for GPR-LA, LAG, LP Type

ERMS (Energy Reduction Maintenance setting)

The ERMS is a function to reduce the arc energy to ensure workers' safety. it is used to reduce the SIG protection settings in order to trip faster when an internal arc fault occurs. It is acknowledged as one of the recognized solutions to improve the protection of works against internal arc-fault caused by faulty electrical equipment, abnormal environmental conditions, lack of maintenance, ingress of mice. The ERMS function, when engaged, allows the A/P type trip unit to operate faster.

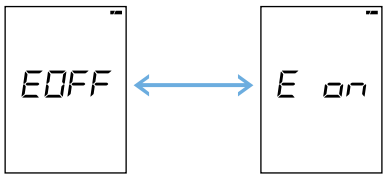
When ERMS is engaged. the corresponding information is made available as follow:
- on the trip unit face with the PTA alarm LED and number 22-23 contact operation on ACB terminal.
- on the trip unit display with 'ERMS' character.
- through the customer communications network (Modbus-RTU)

1. To use the ERMS function, jump between two ends of ERMS terminal.

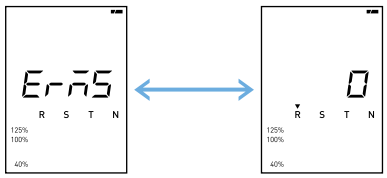
2. Digital input:
   - input (EM1-EM2): Select ERMS.
   ※ All DI should be connected to dry contact.

3. Digital output:
   - Output (22-23) : alarm ERMS activated

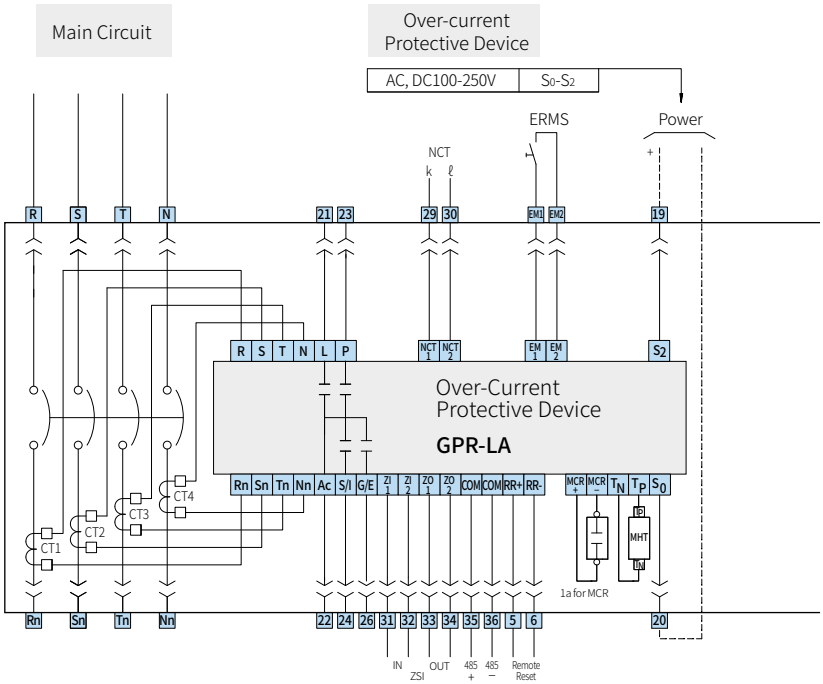
4. It enables ERMS on the trip unit LCD or via user network like Modbus-RTU. The trip unit minimizes S, I, and G protection setting.



ERMS function enable / disable..



when ERMS is enable, display ERMS status.



6.6 Operating Manual for GPR-LA, LAG, LAZ, LP, LH, SA, SP Type

The beginning screen



If control source is applied, each phase load-current is automatically displayed in regular sequence of R-S-T-N. Under the rated 3 %, it marks "0".

In case of pickup setting switch change



In case of pick up setting switch adjustment, setting value is automatically displayed and after turning on back light, it is automatically turned off.

In case of time setting switch change



In case of setting switch adjustment, setting value is automatically displayed and after turning on back light, it is automatically turned off.

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3) Data check method



It explains relay menu and key operation method, setting method through window and example.
By using key panel below, it can use all menu of the relay.

There is a button saving list for checking setting data.

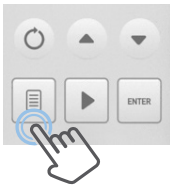
There is a button for moving data.

ENTER — After moving to the item you want to know, there is a button for carrying out.

- List button ( ) - In case of pushing 1 time: Shows maximum current.

×1 time





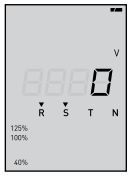
Push list button once.

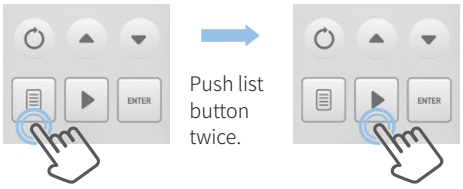
This mode shows phase with maximum current.

- List button ( ) - In case of pushing 2 times: Voltage confirmation (E type only)

×2 times

ENTER





Push list button twice.

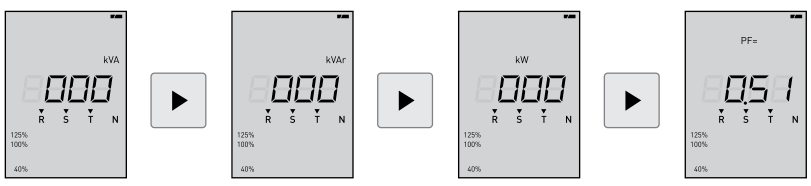
Using movement button, you can confirm voltage.

For E type, pushing list button twice shows measured voltage value.
When it is 3P type, it shows phase-to-phase voltage only. And when it is 4P type, it shows both phase-to-phase voltage and phase voltage. By pushing movement button you can check voltage value which you want.

- List button ( ) - In case of pushing 3 times: Power confirmation (E type only)

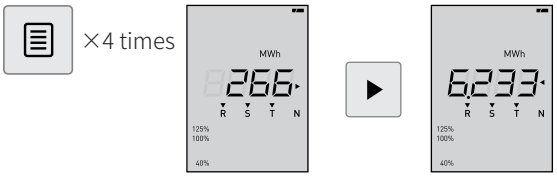
×3 times

ENTER



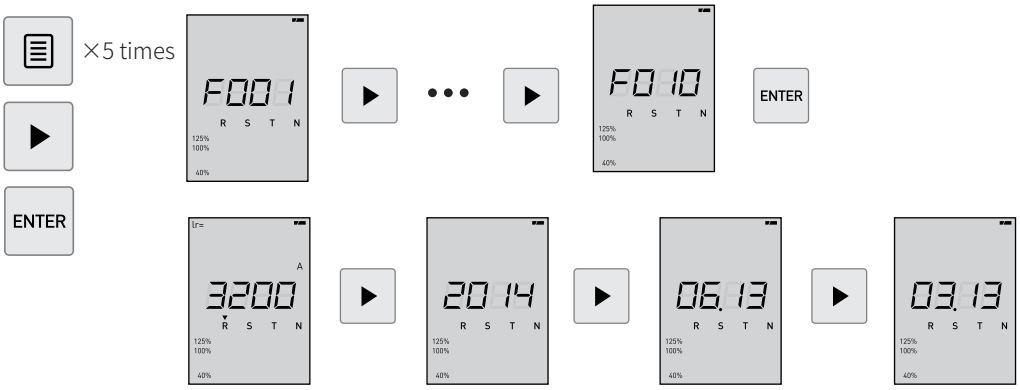
For E type, pushing list buttom 3 times shows measured power value. Each value shows each total for P, Q, S, PF in sequence. In case of negative PF value, positive value will be shown marked with R (reverse).

• List button ( ) - In case of pushing 4 times: Energy



This mode shows active energy value.
Energy is shown in two screens. When there is remaining value, arrow will shown and you can move by movement button.

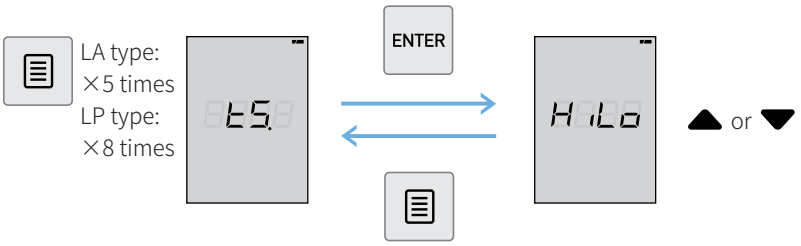
• List button ( ) - In case of pushing 5 times: Fault confirmation



This is list checking accident record. The accident is stored in 10 data at this point in time, F001 is the newest. In case of data number exceeds, it is automatically deleted from the older data. Accidental contents, type-phase load current and accident occurrence time are stored, and you can confirm it by movement button.
Time information is shown in year, month, day, hour, minute in sequence. After choose data with movement button and then push enter button, you can check saved data.

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- List button ( ) - In case of pushing 5 times: Fault confirmation



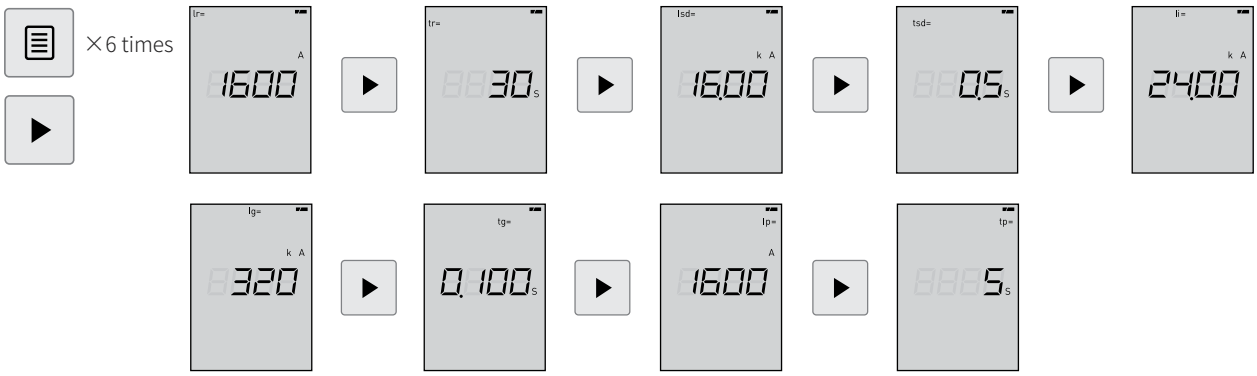
Press the list button on the front of the OCR “n” times until the “tS” menu appears on the display.

In the “tS” menu, press the ENTER button. “HiLo” will appear and the Test function will be activated.

Press the Up or Down arrow button to perform the TEST operation
- Up arrow button : Short-time and Instantaneous test
- Down arrow button : Long-time test.

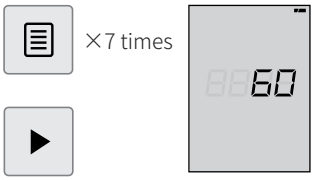
press the menu button to return.

- List button ( ) - In case of pushing 6 times: Set information confirmation.



This mode can check pickup setting value and time setting value per each item.
You can check each menu by movement button in regular sequence of LTD > STD > INST > PTA > GFT > ELT.
Each menu shows its pickup setting value, delayed time value moving by button.

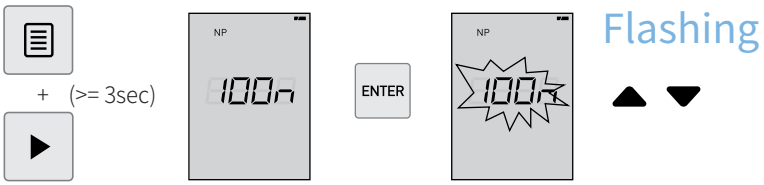
• List button ( ) - In case of pushing 7 times: Frequency confirmation



You can set the frequency value. Frequency is available for 50/60 Hz.

※ When changing frequency, you have to reboot it with power source once disconnected and reconnect it.

• List button ( ) + Movement button ( ) - In case of pushing both for 3 seconds : Neutral protection setting



This is N phase protection. N phase protection can be set as 100 %, 50 %, non of In value.
Once you push enter button, figures flash, it turns into setting mode. Use hi/low button during this setting mode and then press enter button when you finished setting.

• Movement button ( ) - In case of pushing it 1 time: Baud rate setting



Modbus address
1-240

Baud rate
9,600/19,200/38,400
38,400 (default)

Swap
ON, OFF (default)

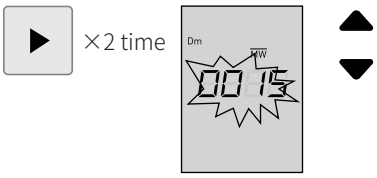
This is communicational parameters setting mode. Pressing enter button at each screen make figures flashing, turning into setting mode. Using hi/low button to adjust value, and press the enter button when you finished setting.

• Communication protocol : Modbus-RTU • Data bit: 8 bit • Parity: non • Stop bit: 1

Modbus address can be set between 1-240, and basically set as 1.
Baud rate can be set as 9,600/19,200/38,400 and basically set as 38,400.
Swap is transmitting function which able to transmit data byte more than word.
Ex) 0×1234 > 0×3412

06 Protection Relay - GPR

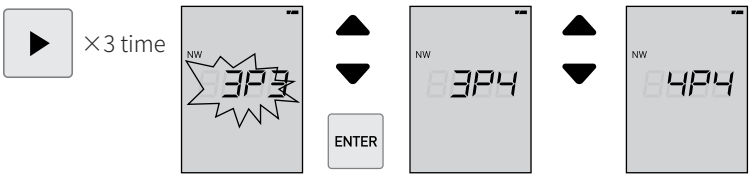
• Movement button (▶) - In case of pushing it twice : Demand time setting (P type only)



Flashing

Demanding value is average value during given time.
Set the time for demanding value calculation of power and current.
Pressing enter button, figures are flashing, it turns into setting mode.
Using hi/low button to adjust value, and press the enter button when you finished setting.
Time setting is available from 5 to 60 minutes, by 1minute unit.
Default setting is 15 minutes.

• Movement button (▶) - In case of pushing it 3 times: Network setting.



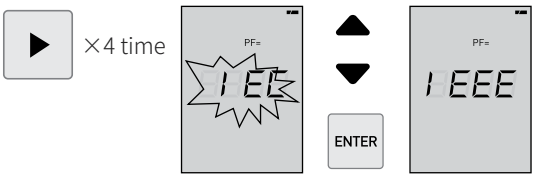
This is setting mode of network type.
It is restricted by type of OCR 3-pole/4-pole.
※ Network setting criteria is determined by type of ACB.

ACB CT	OCR type	Network
3	3-poles	3P3W, 3P4W
4	4-poles	4P4W

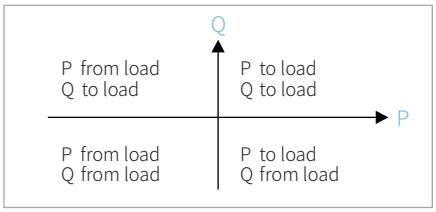
※ In case of 3P ACB but with 4 wire connection, you have to set network as 3P4W.

When you press enter button, figures are flashing, it turns into setting mode. Using hi/low button to adjust value, and press the enter button when you finished setting.

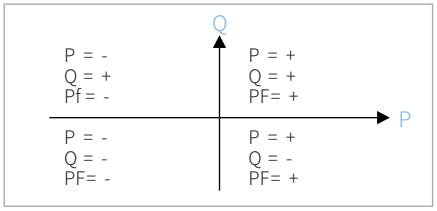
• Movement button (▶) - In case of pushing it 4 times: Power factor sign convention setting (P type only)



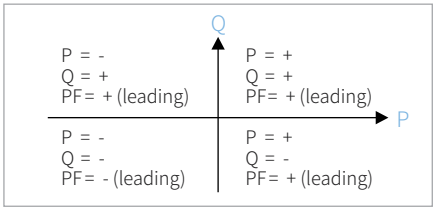
Flow of active and reactive power



IEC

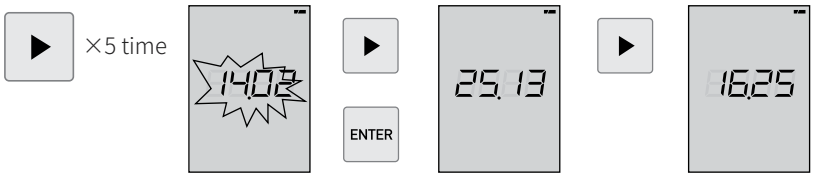


IEEE



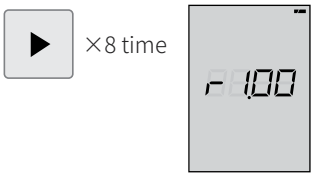
This is setting mode regarding power factor sign convention according to IEC and IEEE.
Signs are determined from the active power and reactive power flow.
Pressing enter button, figures are flashing, it turns into setting mode. Using hi/low button to adjust value, and press the enter button when you finished setting. Basic setting is IEC convention.

• Movement button (▶) - In case of pushing it 5, 6, 7 times: Time setting



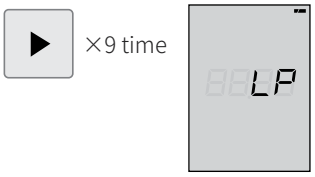
This mode shows and sets the time of your OCR.
Time is expressed in year, month, day, hour, minute in sequence each in two digit number.
Pressing enter button, figures are flashing, turning into setting mode.
Using hi/low button to adjust value, and press the enter button when you finished setting.
※ OCR time runs by replaceable battery. You have to replace it when 'LOW'.

• Movement button (▶) - In case of pushing it 8 times: OCR version confirmation



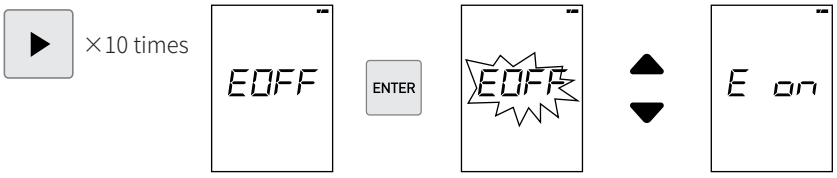
This mode shows version of your OCR.

• Movement button (▶) - In case of pushing it 9 times: OCR model number confirmation



This mode shows model number of your OCR.

• Movement button (▶) - In case of pushing it 9 times: OCR model number confirmation



Press up or down button to use the ERMS function.

E on: enable
Eoff: disable

07 Insulation Resistance Test and Dielectric Withstand Test

The insulation resistance test and dielectric withstand test for main circuit and control circuit are performed as follows:

7.1 Main Circuit

- Dielectric withstand voltage performance is AC 3,500 V for one minute.
- Use a DC 500 V insulation resistance tester (megger) (over 300 ohm).

7.2 Control Circuit (to earth)

- Dielectric withstand voltage performance is AC 1,500 V for one minute.

  For DC 24 V rating of motor charging and closing operation circuits, its characteristic is AC 500 V for one minute (control circuit terminals ①, ②, ③).
  For DC rating of the following control circuits, withstand voltage test is impossible.

  - Control circuit terminals ⑨ and ⑩ of undervoltage trip device (UVT).
  - The terminals ⑲ and ⑳ of GPR protection relay.

- Use DC 500 V insulation resistance tester (megger) for insulation resistance test.

⚠ CAUTION

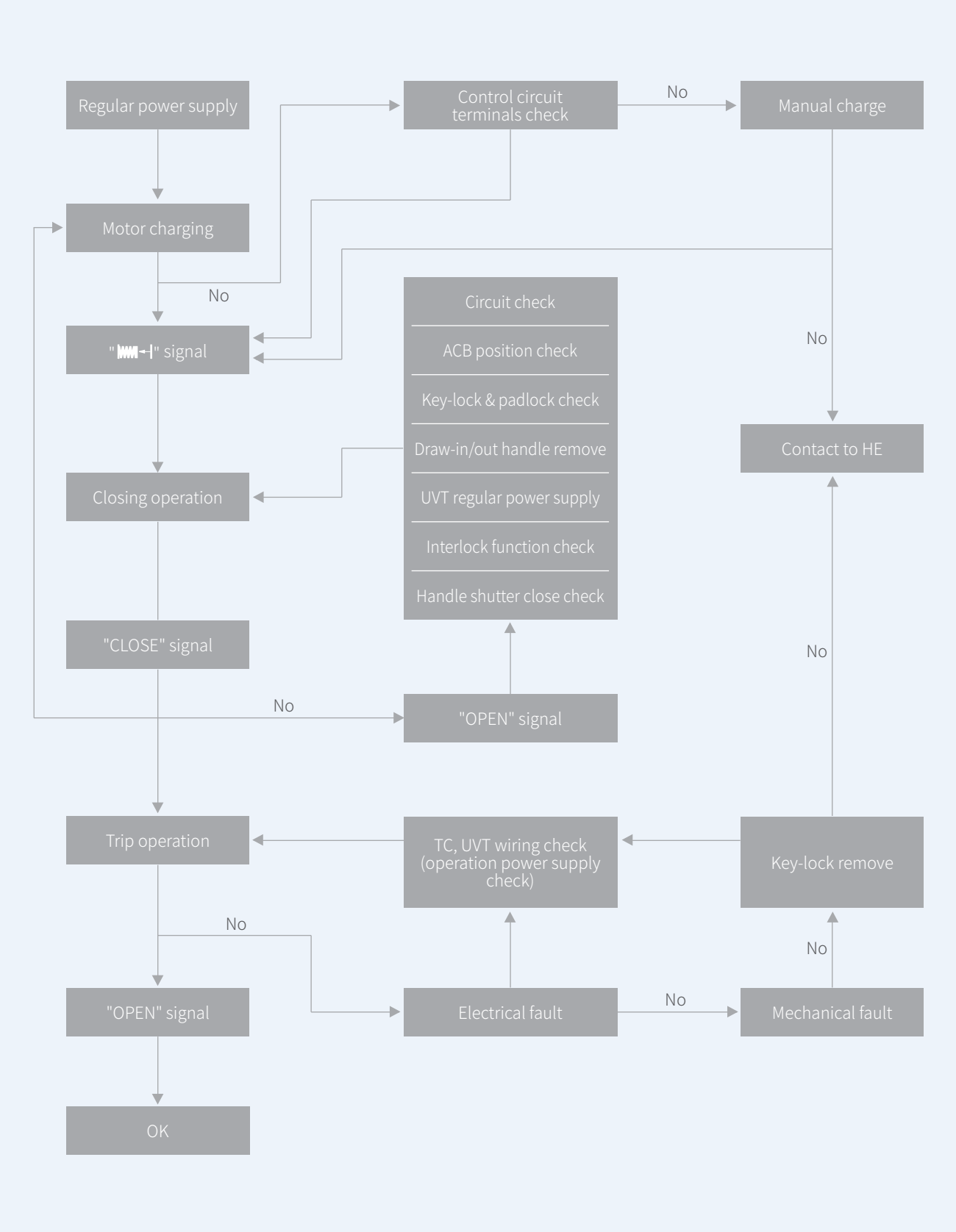
The circuit connected with OCR do not test insulation resistance test and dielectric test.
There are possibilities of malfunction and breakdown by damaging for internal protection device.
(Control Jack : No.19~20, VR, VS, VT, VN)

08 Troubleshooting Points

No.	Trouble	Phenomenon		Expected Causes	Corrective Actions
		Machine	Electrical		
1	Close fault	×	×	1. Mechanism badness	1. Contact HE
				2. Whether it is charging by handle mechanism	2. Contact HE
				3. Is close rod returned ?	3. Close rod returning check
				4. Whether inertia latch is jamming or not ?	4. Inertia latch movement check.
				5. Is ACB placed in proper position (ISO-TEST-CON) ?	5. Move to proper position
				6. Is GPR manual reset button pull out ?	6. Push back GPR manual reset button
				7. Is UVT coil applied proper voltage ?	7. UVT source check
				8. Is position lock button handle cover pushed in ?	8. Handle cover comeback . Push draw in/out handle and adjust position confirm button returning
				9. The breaker caught interlock ?	9. Interlock dissolution
				10. Is trip latch turned down ?	10. Trip latch returning
		×	O	1. Is close rod returned ?	1. Close rod returning
				2. Is trip latch turned down ?	2. Trip latch returning
		O	×	1. Is proper voltage applied to CC coil ?	1. CC coil voltage check
				2. Is CC coil operates normally ?	2. Coil replacement
				3. Is CC coil stroke in proper position ?	3. Stroke check
				4. Is UVT coil applied proper voltage ?	4. UVT source check
				5. Is over current relay reset ?	5. Reset OCR
2	Open fault	×	×	1. Whether any interlock system exist ?	1. Contact HE
				2. Whether trip latch exists bur ?	2. Contact HE
		O	×	1. Whether proper voltage applied to TC coil ?	1. TC coil voltage check
				2. Whether TC coil exists mechanical interference	2. After voltage is applied, check core movement
				3. Whether proper voltage applied to UVT coil ?	3. UVT coil voltage check
3	Charging fault	×	×	1. Whether handle mechanism cam operates normally ?	1. Contact HE
		O	×	1. Whether proper voltage applied to motor ?	1. Voltage check (85-110 % of rated voltage)
				2. Whether motor control unit is damaged	2. Motor output voltage check
				3. Whether motor ratchet (M/B) is proper	3. Part check

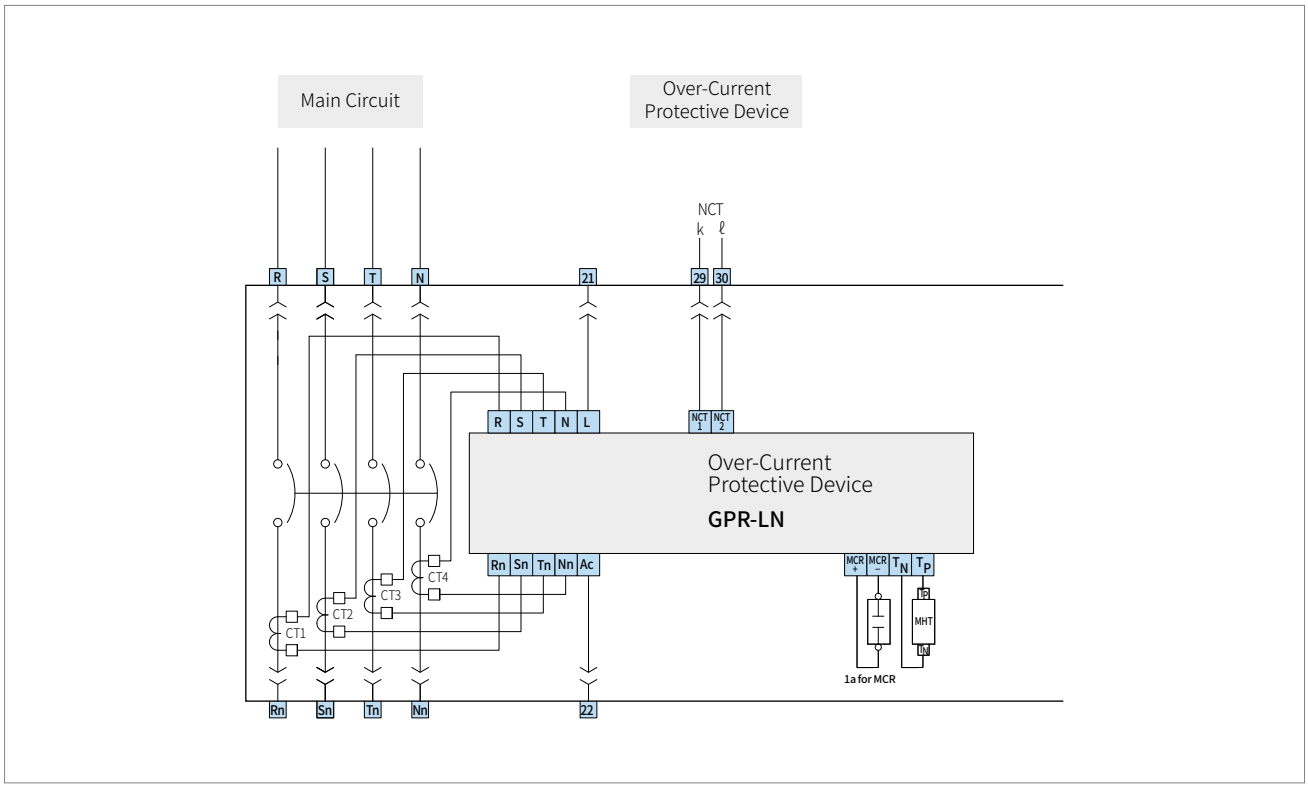
09 Appendix

9.1 Closing and Trip Operation Cycle

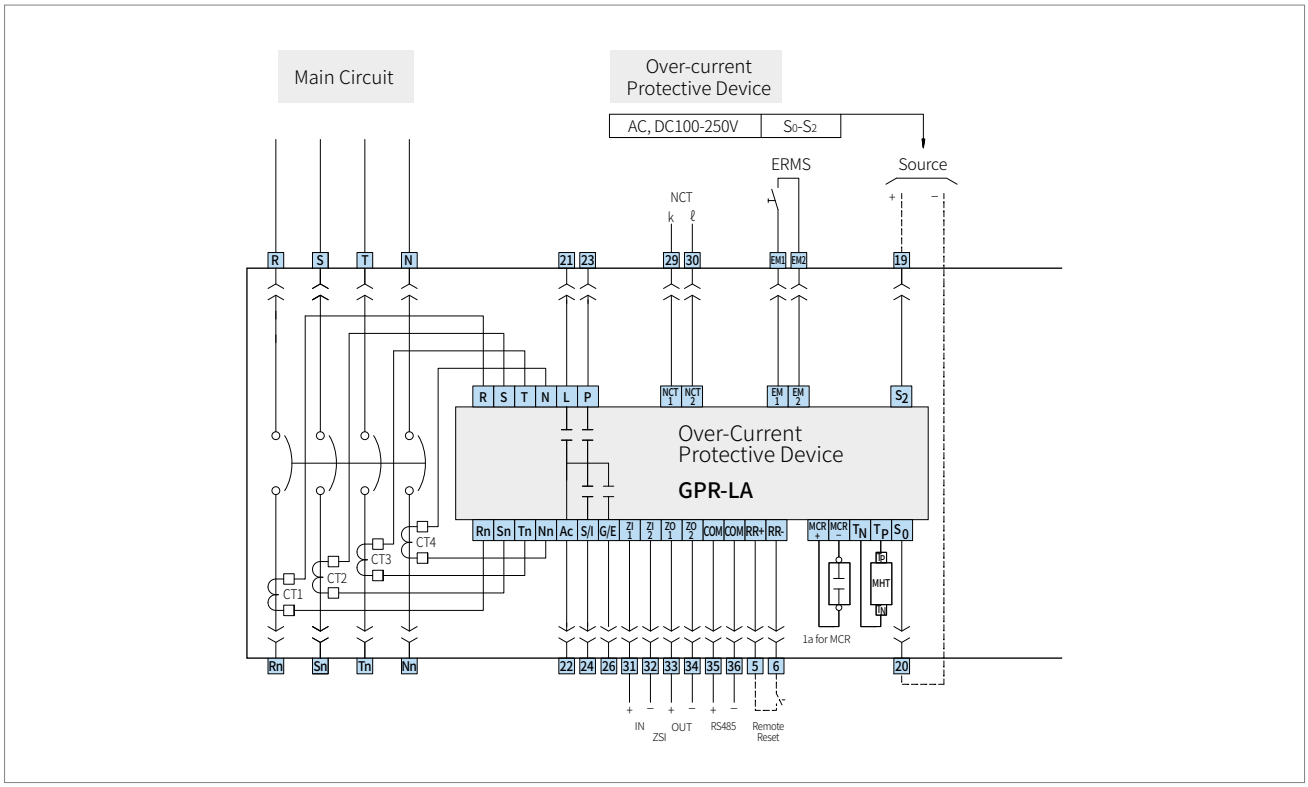


9.2 Wiring Circuit for GPR Protection Relay

GPR-LN

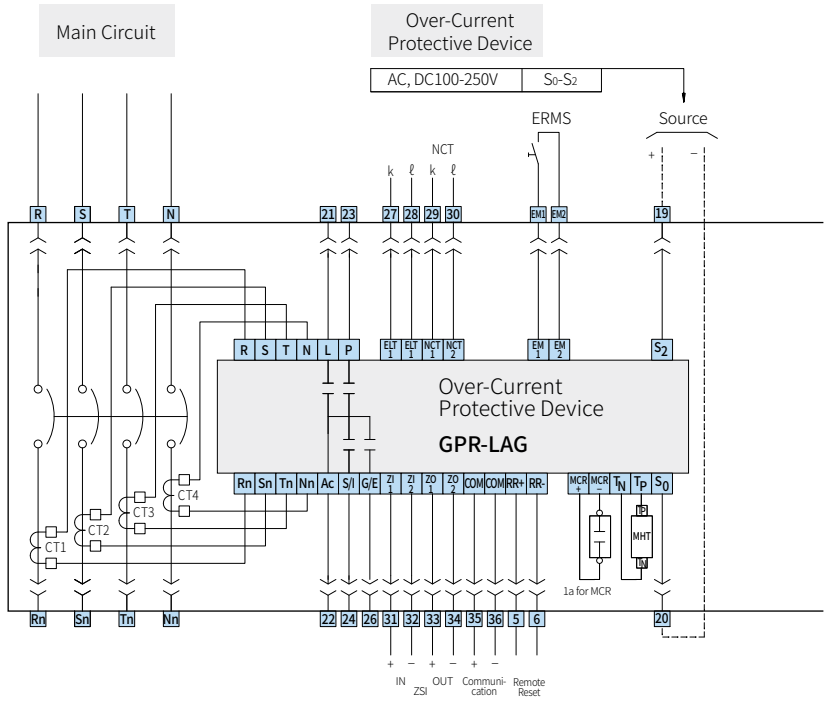


GPR-LA

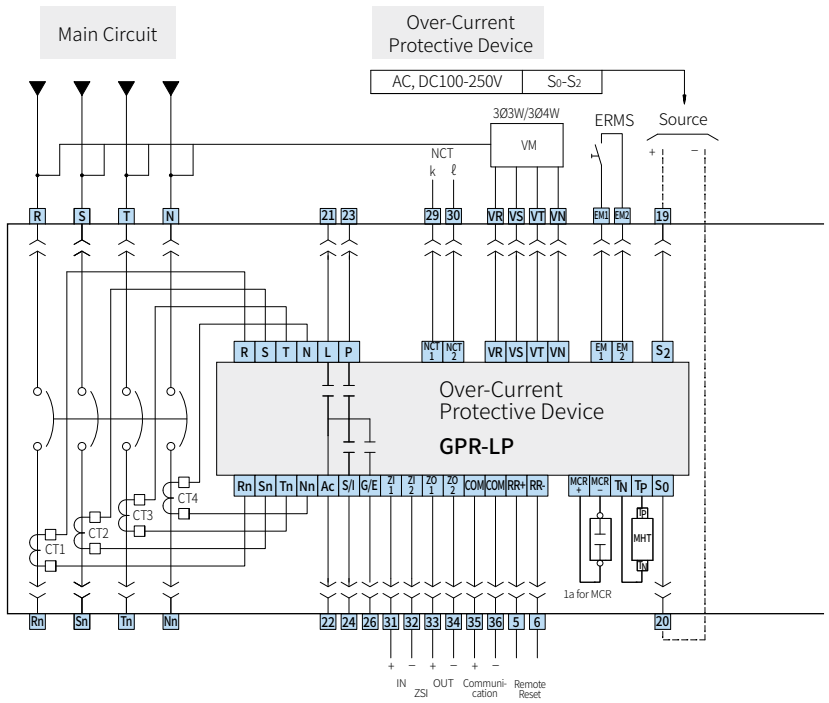


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GPR-LAG

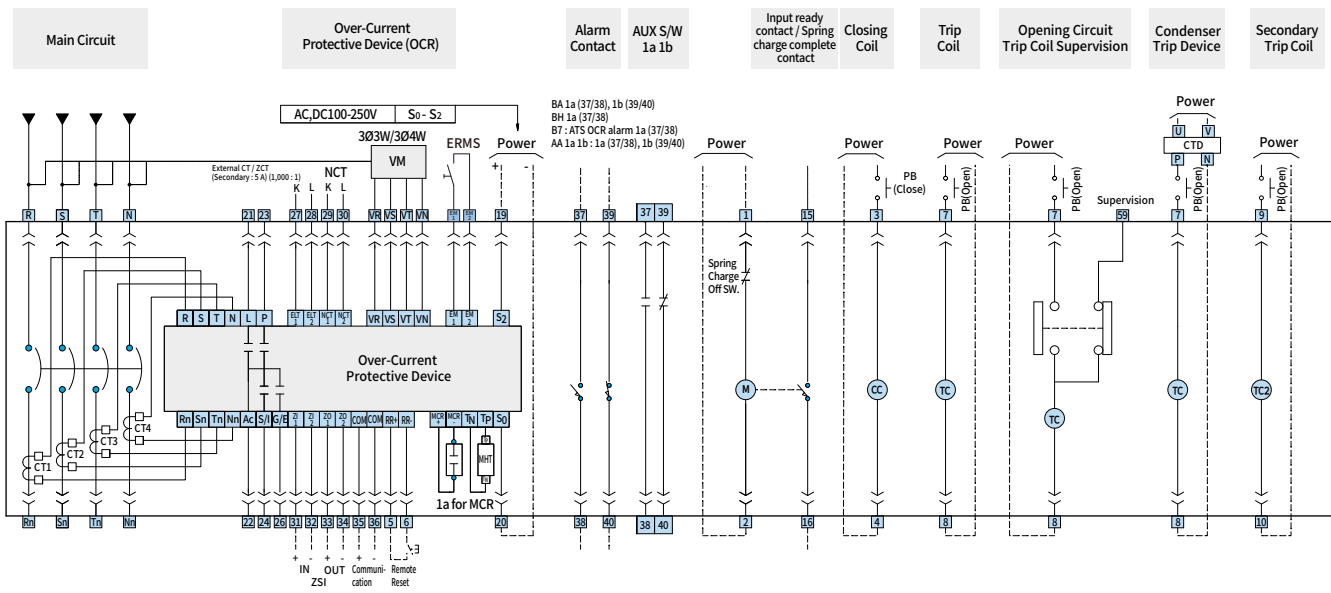


GPR-LP



09 Appendix

9.3 Wiring Circuit for ACB



Symbol description

CT	Current transformer
L	LTD terminal
PT	PTA (pre-trip alarm) terminal
G	Ground fault terminal
S/I	STD/INST terminal
Ac	Common terminal
NCT	Neutral current transformer
ZI	Zone selective input
ZO	Zone selective output
MCR +,-	MCR input terminal
Tp / Tn	MHT input source
M	Charging motor
CC	Close coil (close)
TC	Trip coil (open)
UVT	Under voltade trip coil
CT	Magnetic hold trigger
S0 / S2	GPR protection relay source power
EM1 - EM2	ERMS

Terminal description

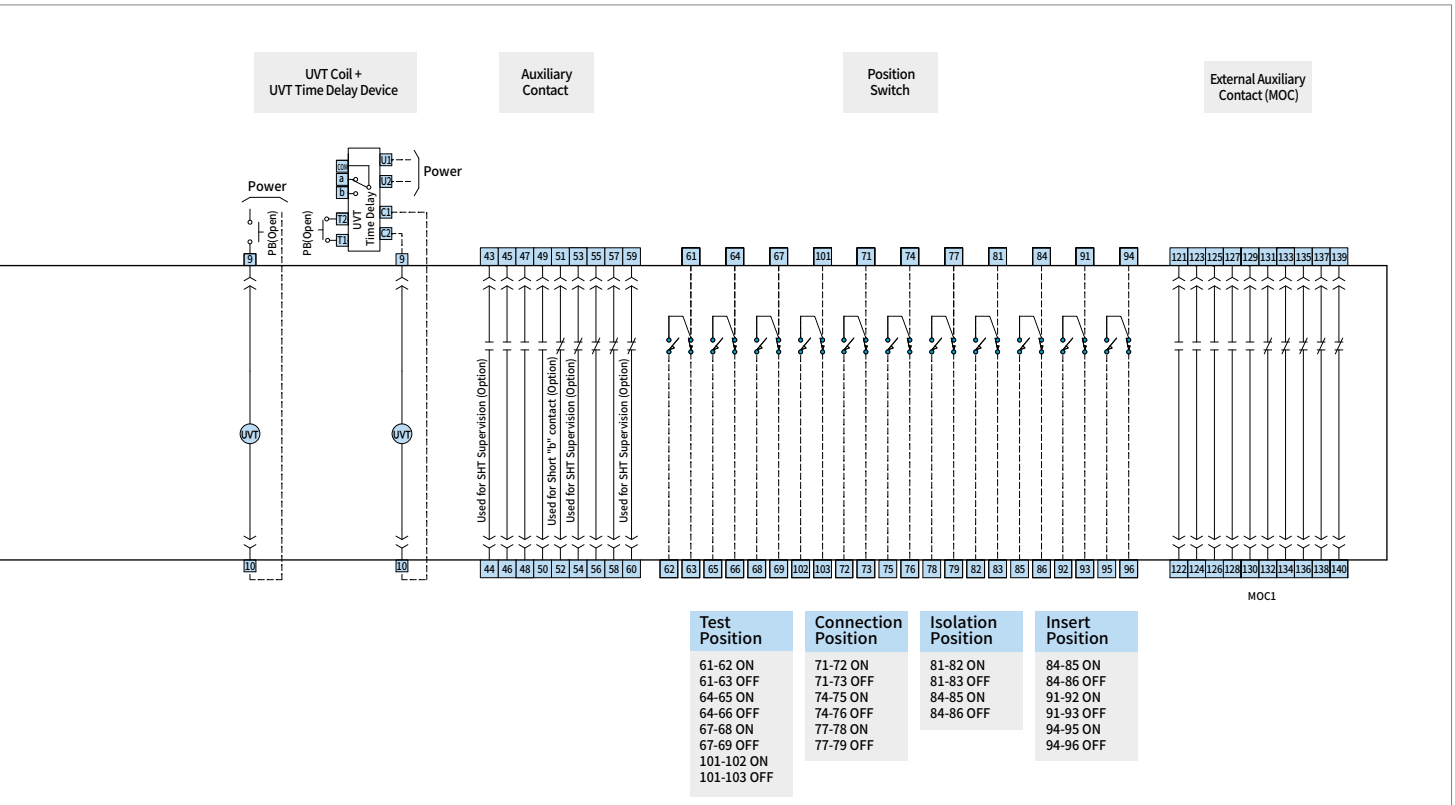
1 2	Charging motor (M) source power
3 4	Closing coil (CC) source power
7 8	Open coil (TC) source power
9 10	UVT coil terminal
15 16	Charge complete contact
19 20	OCR source power
22 21	LTD contact
22 23	PTA/TEMP contact
22 24	STD/INST contact
22 26	GFT/ELT contact
29 30	NCT input terminal
31 - 34	ZSI
41 - 60	AUX. contact
61 - 93	Position switch

—— Manufacturer's wiring

---- User's wiring

Disconnecting device (draw out type)

- RR / Remote Reset
- VM / Voltage Module
- VR~VN / Voltage phase input
- R~N / Current input
- Rn~Nn / Current input



Test Position	Connection Position	Isolation Position	Insert Position
61-62 ON	71-72 ON	81-82 ON	84-85 ON
61-63 OFF	71-73 OFF	81-83 OFF	84-86 OFF
64-65 ON	74-75 ON	84-85 ON	91-92 ON
64-66 OFF	74-76 OFF	84-86 OFF	91-93 OFF
67-68 ON	77-78 ON		94-95 ON
67-69 OFF	77-79 OFF		94-96 OFF
101-102 ON			
101-103 OFF			

Control jack lay-out

OCR & OCGR														CONTROL CIRCUIT					AUXILIARY SW.									
	POW	OCR CONTACT					RR	NCT	ZSI-IN	ZSI-OUT	ERMS	RS485	BA(BH)/AA	VM/ZCT	M	CC	TV/TCS	UVT/TC2	CHA	1a	2a	3a	4a	1b	2b	3b	4b	5b	
		COM-22	L-21	S/I-24	PTA-23	G/E-26																							
	+19	21	23			5	+29	+31	+33	EM1	+35	37	39	VR/27 VT	1	3	7	9	15	43	45	47	49	51	53	55	57	59	
-20	22	24	26		6	-30	-32	-34	EM2	-36	38	40	VS/28	VN	2	4	8	10	16	44	46	48	50	52	54	56	58	60	
OCR Protection relay															Operating circuit					Auxiliary switch									

OCR & OCGR															
	POW	OCR CONTACT					RR	NCT	ZSI-IN	ZSI-OUT	ERMS	RS485	BA(BH)/AA		VM/ZCT	
		COM-22	L-21	S/I-24	PTA-23	G/E-26										
	+19	21	23				5	+29	+31	+33	EM1	+35	37	39	VR/27	VT
-20	22	24	26			6	-30	-32	-34	EM2	-36	38	40		VS/28	VN

- GPR Protection relay -

CONTROL CIRCUIT				
M	CC	TV/TCS	UVT/TC2	CHA
1	3	7	9	15
2	4	8	10	16

- Operating circuit -

AUXILIARY SW.								
1a	2a	3a	4a	1b	2b	3b	4b	5b
43	45	47	49	51	53	55	57	59
44	46	48	50	52	54	56	58	60

- Auxiliary switch -

Essential for Today
Potential for Tomorrow

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Cheongju Distribution Campus	20, Guksasandan 1-ro, Oksan-myeon, Heungdeok-gu, Cheongju-si, Chungcheongbuk-do, Korea		
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